No-regret Learning and a Mechanism for Distributed Convex Optimisation and Coordination

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Abstract

We develop a novel indirect mechanism for coordinated, distributed multi-agent optimisation, planning and decision-making. We consider coordination problems that can be stated as a collection of single-agent convex programmes coupled by common soft constraints. A key idea is to recast the joint optimisation problem as distributed learning in a repeated game between the original agents and a newly introduced group of *adversarial* agents who influence *prices* for decisions and facilitate coordination. The adversarial agents benefit from *arbitrage*: that is, their incentive is to uncover interaction-constraint violations and, by selfishly adjusting prices, encourage the original agents to avoid making decisions that cause such violations. If all agents incur sub-linear regret in the course of this repeated interaction, we are able to show that our mechanism can achieve design goals such as social optimality (efficiency) and Nash-equilibrium convergence to within an error which approaches zero as the agents gain experience. Therefore, these desirable coordination outcomes are guaranteed if all agents employ no-regret learning algorithms, regardless of whether learning takes place in the full- or in the partial-information setting. Alternatively, these bounds can also model non-algorithmic agents who behave sub-rational to a degree that can be bounded by a sub-linearly growing regret bound. Our error bounds are deterministic or probabilistic, depending on the nature of the agents' regret bounds. In addition to our theoretical guarantees, we illustrate our method in several application domains. These include network routing, a simple market of non-atomic goods, as well as a pollution reduction scenario where our adversarial agents learn how to set emission taxes.

1 Introduction

Planning is a process that leads to a decision on an action or a policy with the aim of serving an objective. Many important planning and decision-making tasks can be addressed by convex optimisation. Examples can be found among resource allocation problems, in source routing, and in model-predictive control. Distributed approaches come with many benefits including preservation of information privacy, a reduction in the maximum computational requirements of any particular agent, and robustness to agent failure [SD99]. In this work we provide a distributed approach to multi-agent planning in the presence of soft, convex inter-agent constraints.

In situations where multiple agents interact in a common environment agents cannot make decisions independently but need to be coordinated. That is, their optimisation problems are coupled by interaction constraints. When these interaction constraints are convex in the decisions of the agents, one can achieve coordination by solving a large, joint convex programme. However, several issues preclude this centralised approach from being implemented.

One issue is that multi-agent systems are not always made up of high-spec, electronic devices, and centralised computation of the optimal solution can be simply *intractable*, due to restrictions on the computational ability of the agents, or on their communication capabilities. These restrictions can be interpreted as bounded rationality of agents, or as physical restrictions imposed by their environment.

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Furthermore, in order to state the centralised optimisation problem, all agents have to report *truthfully* their *type*, i.e. all relevant information including their utilities. As an agent's type is private and identifying information, this renders the centralisation implausible in the presence of competing agents who fear a competitive disadvantage from revealing their type, or strategic agents who might choose to misreport their type for personal gain.

A large body of work seeks to address these issues across a host of fields, including theoretical computer science, robotics, artificial intelligence, control and economics. In the engineering-related disciplines emphasis is put on the design of algorithmic solutions that can operate in a distributed manner, while less consideration is given to agents that are not obedient and might chose to deviate from the prescribed protocol. In contrast, self-interested agents are the focus of attention for the economic theory of mechanism design. Works from this area aim to design and control the interaction of rational, self-interested agents in a way that leads to equilibria with desired system wide properties. However, most economic mechanisms suffer from either overly simplistic model assumptions, such as assuming no preference interactions between the goods in a market, or on perfect rationality of agents achieved by unbounded computation, as in the VCG mechanism applied to combinatorial auctions. So, even if they show that the mechanism coincides with a game with desirable equilibria, it is not clear how the agents might efficiently achieve this equilibria in a computationally tractable manner.

In providing a mechanism for distributed optimisation, our work is applicable to distributed model-predictive control [NM14] and planning. However, it also sits within the emerging field of (distributed) computational mechanism design [DJP03]. This field aims to bridge the gap between the economic perspective of mechanism design and the engineering-oriented aim of computational tractability. In particular, we derive a mechanism that achieves an approximately socially optimal coordination outcome over the course of a repeated game playing interaction. The approximation bounds we give depend the model parameters, i.e. the number of agents, interactions, and game repetitions, as well as on bounded rationality of the participating agents through regret bounds of their strategies and behaviours. Therefore, we can view our results as a quantification of the trade-off between social optimality, problem complexity and bounded rationality of the agents.

Our approach coordinates the different, individual goals of the participating *player* agents, $P_1, \ldots, P_k$, to achieve a globally desirable plan. We consider multi-agent planning problems whose optimal outcomes can be stated as k single-agent convex optimisation problems that are coupled by n linear, soft constraints. By a *soft* constraint, we mean that violations are feasible, but are penalised by a convex loss function such as a hinge loss. This representation includes many application domains. For example, in network routing problems, each agent's feasible region represents the set of paths from a source to a sink, its objective is to find a low-latency path, and a soft constraint represents the additional delay caused by congestion on a link used by multiple agents. More generally, we can think of the the constraints as setting penalties for over-consumption of n reasources that are commonly shared by the player agents and consumed by their plans.

Since all coupling *inter-agent constraints* are soft, each agent's feasible region does not depend on the actions of the other agents.[1] Hence, the agents can plan independently, and still guarantee that their joint actions are feasible. We can view this interaction via soft-constraints as a *convex game* [SL07]: each player simultaneously selects her plan from her convex feasible set, and if we hold all plans but one fixed, the remaining player's loss is a convex function of her plan. By playing this convex game repeatedly, the players can learn about one another's behaviour and adjust their actions accordingly. By employing appropriate learning algorithms they can ensure convergence to an equilibrium [BHLR07, GGMZ07]. Unfortunately, while distributed and robust, this naïve setup can lead to highly suboptimal global behaviour. For example, in the network routing setting, a selfish agent which can gain any benefit from using a congested link will do so, even if the resulting cost to other agents would far outweigh its own gain [Rou07].

To overcome this problem, we introduce as a key concept additional *adversarial* agents $A_1, ..., A_n$ that control variables of a dual optimisation problem. Interpreting the agents' plans as consuming resources each of these adversarial agents collect additional usage fees, which they control. Like the player agents, the new agents are self-interested, but, collectively, they encourage the player agents to avoid excessive resource usage. In particular, we show below how to define their revenue functions so that they effectively perform *arbitrage*, allocating extra constraint-violation costs to each player agent in proportion to its responsibility for the violations.

[1] For a treatment on how our mechanism can be used in settings where inter-agent constraints are hard refer to Appendix C.

The adversarial agents' revenues and payments are designed to *decouple* the individual agent planning problems. Indeed, a player agent only perceives the actions of other player agents through their effects on the choices of the adversarial agents. So, under our mechanism, the player agents never need to communicate with one another directly, nor do they need to communicate their type. Instead, they communicate only the relevant information with the adversarial agents to receive price information. Moreover, if player agent never affects a particular interaction-constraint, it never needs to send messages to the corresponding adversarial agent. This decoupling can greatly reduce communication requirements and increase robustness compared to the centralised solution in which a central planner must communicate with every agent.

Having decoupled the individual planning problems each agent no longer needs to worry about the whole planning problem. Instead, they need only to solve a local, online convex problem (OCP) [Gor99, Zin03] independently and selfishly. To solve their local OCPs, the agents could use any learning algorithm it desired, but, in this paper, we explore what happens if the agents use learning algorithms to update their decisions under various types of feedback, including bandit feedback, and gradient feedback. And, if these decisions incur sublinear-regret various desirable system-wide behaviours ensue.

Specifically, if each agent's average per-iteration regret approaches zero, the agents will learn a globally optimal plan, in two senses. First, the *average per-iteration cost*, summed over all the agents, will converge to the optimal social cost for the original planning problem without the adversarial agents. Indeed the approximation error in the social cost after a finite number of iterations can be bounded in terms of the average regrets of the agents. Second, the *average overall plan* of the player agents will converge to a socially optimal solution of the original planning problem.

Our optimality guarantees hold true for any type of agent decision-making that happens to incur sub-linear regret (and therefore vanishing average regret) with sufficiently high confidence. In this sense, the sub-linear regret requirement is less stringent than the assumption of perfect rationality typically made in economic contexts; but, it does impose a restriction on the agents' degree of sub-rationality during their interaction. From an algorithmic perspective, fortunately, the agents can always choose from a host of existing no-regret algorithms to guarantee they indeed meet the desirable sub-linear -regret requirement.

When the agents receive very limited, or merely bandit information about the costs incurred by their past decisions, no-regret guarantees of learning algorithms are typically probabilistic statements. For such cases, high-probability bounds on the approximation error in the social cost are possible, and there exist commonly satisfied conditions under which guarantees of convergence of the plans also hold with probability 1.

Our results lead us to propose two different mechanism variants, namely an *online* set-up and a *negotiation* set-up. In the former, motivated by the convergence results, learning takes place online in the classic sense. All agents choose and execute their plans and make and receive payments in every learning iteration. In the latter, motivated by the error bounds on the social cost, the agents learn and plan as usual, but only simulate the execution of their chosen joint plan. The simulated results (costs or rewards) from this proposed joint plan provide feedback for the learners, and after a sufficient number of learning iterations, each agent averages together all of its proposed plans and executes the average plan.

Both set-ups, but in particular the negotiation set-up, can be interpreted as a very simple market. Here we might interpret a plan as a desired intention to buy or sell resources and an interaction-constraint as a rule specifying the extent to which buy and sale levels need to match. The adversarial agents can be interpreted as arbitrage agents that facilitate coordination by negotiating the right resource prices.

Just as with any mechanism, because our agents are selfish, we need to consider the impact of individual incentives. Focusing on the negotiation setup, we provide (mainly asymptotic) guarantees of Nash-equilibrium convergence of the overall learning outcome as well as classic mechanism design goals such as efficiency.

Our work extends and generalises research that first appeared in the proceedings of the *7th International Conference of Autonomous Agents and Multiagent Systems (AAMAS 2008)* [CG08b]. In contrast to our earlier work, our new results rest on the assumption of probabilistic regret statements. This facilitates settings, where the agents have to learn how to plan in the partial information setting. That is, in each stage of the repeated interaction, the agents are allowed to observe only the cost of the last action rather than the full cost function of all possible actions.

This is useful in settings where the agents cannot be assumed to possess complete understanding of the cost effects of all their decisions and where computing or communicating this type of information is infeasible or undesirable.

Furthermore, the previous work assumed that the agents were restricted to consume resources. In contrast, we make less stringent assumptions on the functional relationship between plans and interaction-constraints which allows us to tackle a more general class of distributed convex optimisation problems. As a result, we are also able to consider a wider range of application domains such as full markets with buyers and sellers as well as pollutant emission control.

The structure of the remainder of this paper is organised as follows: Sec. 2 reviews some algorithmic concepts such as OCPs and no-regret algorithms. It then continues by introducing the model assumptions of the overall planning problems considered, establishes some notation used throughout the rest of the paper and states the coordination objective and the social choice function to be realised by our mechanism.

Sec. 3 describes our mechanism and how it is derived by recasting the original coordination problem as learning in a repeated market game in which the agents adaptively negotiate plans and prices for their plans.

Having described the mechanism, Sec. 4 provides an analysis of its theoretical properties. In particular, we provide (sub-) optimality bounds on the coordination success as a function of the duration of the interaction time, the regret bounds of the learning algorithms and the complexity of the coordination problem.

We also give some bounds on the number of coordination rounds required to ensure the deviation from the suboptimal outcome lies within an ε-ball of the optimal social cost. We also comment on how this bound scales with the number of agents and the regret bounds of their learning algorithms (if they choose to employ no-regret algorithms to update their decisions). Sec. 5 illustrates the viability of our mechanism by applying it to three coordination scenarios. Sec. 6 discusses our work in the context of related work.

2 Preliminaries

As they are integral to both the analysis and implementation of our approach, we will briefly review in this section the notions *online convex programming problems (OCPs)* and *external regret*. Once we have established these notions we go on to describe the multi-agent setting in detail.

2.1 Online convex programming and no-regret algorithms

We assume throughout this subsection that F is a convex subset of a d-dimensional, $\mathbb{R}$-Hilbert space, V, endowed with an inner product, $\langle \cdot, \cdot \rangle_V : H \times H \to \mathbb{R}$, and associated metric $d_V(\cdot, \cdot) : H \times H \to \mathbb{R}$.

In an *online convex program* [Gor99, Zin03], an arbitrary sequence of (stage) *cost*, or *loss*, functions, $(\Gamma_t)_{t \in \mathbb{N}}$, convex on the domain F, is revealed step by step.[2] At each stage t, the OCP algorithm must choose an *action* x_t from the convex *feasible set* F having access to information $\mathbb{I}_t$ about past actions and cost functions up to stage $t-1$. After the choice is made, information $\mathbb{F}_t$ about the current cost function Γ_t is revealed, and the algorithm suffers a cost amounting to $\Gamma_t(x_t)$. The information that the OCP algorithm can use to choose the action x_t is summarised in the information set

$$\mathbb{I}_t = \left\{ \left(x_q \right)_{q<t}, \left(\mathbb{F}_q \right)_{q<t} \right\}.$$

OCP problems can arise in varying setups depending on the kind of information available to the algorithm at the time of decision making and the nature of the *cost feedback* it receives after having made the decision. In the *full-information* model, after having made a decision x_t in stage $t \in \mathbb{N}$, the learner receives the full (stage) cost function as cost feedback, i.e., $\mathbb{F}_t = \Gamma_t$, and the information set is the sequence of past actions and cost *functions*.

By contrast, in the *partial-information* setting, the algorithm is not informed about the entire cost function Γ_t after choosing an action. For example, some gradient oracle may be able to provide the information $\mathbb{F}_t = \nabla\Gamma_t(x_t)$

[2] Equivalently, one could substitute concave reward functions.

instead, and we call this *gradient feedback*. In the most extreme setting only the *cost* of the action chosen is given as feedback, i.e., $\mathbb{F}_t = \Gamma_t(x_t)$, and we call this *bandit feedback*.

To measure the performance of an OCP algorithm, we can compare its accumulated cost up to time step T to an estimate of the best cost attainable against the sequence $(\Gamma_t)_{t=1}^T$. In particular, we estimate the best attainable cost as the cost of the best constant action choice

$$s_T \in \text{argmin}_{s \in F} \sum_{t=1}^{T} \Gamma_t(s)$$

chosen with knowledge of the entire sequence $\Gamma_1, \ldots, \Gamma_T$. This choice leads to a measure called *external regret* or just *regret*:

$$\mathcal{R}(T) = \sum_{t=1}^{T} \Gamma_{(t)}(x_t) - \sum_{t=1}^{T} \Gamma_t(s_T).$$

An algorithm is said to be *no-(external)-regret* if it guarantees that $\max\big(0, \mathcal{R}(T)\big)$ grows slower than $\mathcal{O}(T)$. That is, if there exists a nonnegative sublinear function $\Delta(T) \in o(T)$ with $\mathcal{R}(T) \leq \Delta(T)$. The term *no-regret* is motivated by the fact that the limiting average nonnegative regret of a no-regret algorithm vanishes, i.e., $\limsup_{T\to\infty} \max(0, \mathcal{R}(T))/T = 0$.

Remark 2.1. Note, our definition of a no-regret algorithm differs somewhat from a standard definition that calls an algorithm no-regret if it guarantees that its regret $R(T)$ is in $o(T)$. By that alternative definition, an optimal response agent that always plays the minimum of the stage cost function is not no-regret, whereas by ours it is. The reason is that by the alternative definition, it would be possible to construct sequences of cost functions where the best response agent always incurs negative regret whose absolute value grows linearly (and hence its regret is not in $o(T)$). Since our notion of no-regret algorithm is more general than the alternative one, all algorithms that have been proven to be no-regret under the alternative notion also are no-regret under our definition.

Frequently, and especially for the bandit feedback models, algorithms utilise randomisation. Consequently their actions and hence, regrets, are random variables. In these cases, no-regret guarantees are probabilistic statements. For such no-regret algorithms one can often provide a sublinear bounding function, $\Delta(T, \delta) \in o(T)$, such that, for all $\delta \in (0, 1)$

$$\Pr\left(\mathcal{R}(T) \leq \Delta(T, \delta)\right) > 1 - \delta. \tag{1}$$

The deterministic no-regret case, for which algorithms are currently limited to the gradient feedback setting, arises as the special case where it is also possible to give such a sub-linear regret bound for $\delta = 0$.

2.1.1 Converting bounds on the expected regret into probabilistic confidence bounds

It should be noted that for many algorithms in the literature only a bound on the expected regret is provided. That is, they come with a guarantee of the form $\langle \mathcal{R}(T) \rangle \leq \Delta(T) \in o(T)$. However, we would like to point out that such a bound on the expected regret can be converted into a probabilistic bound as follows: We can always choose a monotonically increasing function f with $f(T) \in o(T)$ such that also $\Delta(T) \in o(f(T))$. Appealing to Markov's inequality it follows that

$$\Pr[\mathcal{R}(T) \geq f(T)] \leq \frac{\langle \mathcal{R}(T) \rangle}{f(T)} \leq \frac{\Delta(T)}{f(T)} = o(T). \tag{2}$$

For example, for $\Delta(T) = k\sqrt{T} + c$ we could choose $f(T) := kT^{\frac{2}{3}}$ and obtain a regret bound:

$$\Pr[\mathcal{R}(T) \leq kT^{\frac{2}{3}}] \geq 1 - T^{-\frac{1}{6}}.$$

Algorithm 1 Online Projected Gradient Descent (OPGD) – "Greedy Projection"

1: **Input:** Feasible set F, initial action $x_1 \in F$, learning rate η, horizon T
2: **for** $t = 1, \dots, T$ **do**
3: Choose action x_t and receive gradient $g_t = \nabla \Gamma_t(x_t)$
4: Set $x_{t+1} = P_F(x_t - \eta g_t)$, where P_F is the orthogonal projection onto F

2.1.2 Algorithms for gradient feedback

Finding no-regret algorithms for the OCP problem is still an extremely active area of research [Bub14, HL16, AHR12, BDKP15, HLGS16]. A popular approach is to assume the presence of a gradient oracle for the loss functions and adapt gradient descent. In particular, it is known that Online Projected Gradient Descent (OPGD), as shown in Algorithm 1 achieves sub-linear regret:

Theorem 2.2. *[Zin03] Let F be a closed, bounded, and non-empty feasible set, let $T \in \mathbb{N}$, and let $\Gamma_1, \dots, \Gamma_T$ be a sequence functions that are convex and differentiable on F. Then applying Projected Gradient Descent and choosing $\eta_t = t^{-\frac{1}{2}}$ results in a regret of at worst*

$$\mathcal{R}(T) \leq \frac{D}{2}\sqrt{T} + L^2\left(\sqrt{T} - \frac{1}{2}\right),$$

where $D := \max_{x,y \in F} d_V(x, y)$ and $L = \max_{x \in F, t \in 1,\dots,T}\{\nabla \Gamma_t(x)\}$. In particular, OPGD is a no-regret *algorithm.*

Since, we can set the learning rate, $\eta = t^{-\frac{1}{2}}$, independently of the horizon, T, and still give a sub-linear regret bound, we call OPGD under this learning rate an *any time*, no-regret algorithm. This important property allows it to be directly applied to both our online and negotiation set-ups without alteration.

2.1.3 Algorithms for the bandit feedback

Another variant of gradient descent is a popular and well studied approach to the bandit-feedback setting. Since here it is not be possible to observe or compute gradients of the cost function by assumption, the key to providing an algorithm based on gradient descent is to derive a one-sample estimator of the gradient of the function. Letting u be a uniformly randomly distributed vector on the unit sphere, then

$$\tilde{g}_t(x) := \frac{d}{\delta}\Gamma_t(x + \alpha u)u \tag{3}$$

is an estimator of the gradient of the function Γ_t at x, smoothed over a ball of size α, centred at x. In particular,

$$E_{u \sim \mathrm{Unif}(S^d)}(g_t(x)) = \nabla E_{v \sim \mathrm{Unif}(B^d)}(\Gamma_t(x + \alpha v))\text{[Bub14]},$$

where S^d and B^d are the d-dimensional unit sphere and ball in the Hilbert space H, respectively, and $\mathrm{Unif}(X)$ denotes the uniform distribution on the set X. We refer to the algorithm using this estimator as Online Projected Stochastic Gradient Descent (OPSGD), and give some pseudocode for it in Algorithm 2.

Theorem 2.3. *[Bub14] Suppose that $F \subset H$ is a closed, convex set, such that $rB^d \subset F \subset RB^d$, for some $r, R > 0$. Let $\Gamma_1, \dots, \Gamma_T$ be a sequence of L-Lipschitz, convex, differentiable functions on F, whose values are bounded by C. Fix α, and assume that OPSGD is run on $(1 - \alpha/r)F$ with learning rate $\eta_t = \eta > 0$. Then, choosing*

$$\alpha = \frac{1}{(2T)^{\frac{1}{4}}}\sqrt{\frac{RdL}{(3 + R/r)C}} \text{ and } \eta = \frac{1}{(2T)^{\frac{3}{4}}}\sqrt{\frac{R^3}{dL(3 + R/r)C}},$$

we have

$$E[\mathcal{R}(T)] \leq 4\left(RdC\left(3 + \frac{R}{r}\right)L\right)^{\frac{1}{2}} T^{\frac{3}{4}}$$

Algorithm 2 Online Projected Stochastic Gradient Descent (OPSGD)

Input: Feasible set F, initial estimate $y_1 \in F$, learning rate η, perturbation parameter α, horizon T
for $t = 1, \ldots, T$ **do**
  Generate a random vector u_t from the uniform distribution on the unit sphere
  Choose action $x_t = y_t + \alpha u_t$ and receive cost $\Gamma_t(x_t)$
  Construct gradient estimator $\tilde{g}_t$ as in (3)
  Set $x_{t+1} = P_F(x_t - \eta \tilde{g}_t)$, where P_F is the orthogonal projection onto F

In combination with the argument in Section 2.1.1 above, this can be used to give a high-probability guarantee of the form:

$$\Pr\left(\mathcal{R}(T) \leq 4\left(RdC\left(3 + \frac{R}{r}\right)L\right)^{\frac{1}{2}} T^{\frac{4}{5}}\right) \leq 1 - T^{-\frac{1}{20}}.$$

In recent work [HL16], an algorithm has been proposed that guarantees

$$\mathcal{R}(T) \leq 2^{d^4}(\log t)^{2d} log(\delta^{-1})\sqrt{T}, \text{ with probability } 1-\delta. \tag{4}$$

While this gives a constructive proof that it is possible to achieve a regret of $O(\sqrt{T})$, it has a particularly unsatisfactory exponential dependence on the dimension d. Exactly what the optimal dependence on the parameters d, and T are is an open question in this research area.

2.2 Multi-Agent Coordination Problem and Notation

We wish to model interaction among k *player* agents, $P_1, \ldots, P_k$. We represent player P_i's individual planning problem as a convex program: choose a vector p_i from a compact, convex feasible set F_{P_i} to minimize the *intrinsic* cost $c_i(p_i)$. Here, $c_i : F_{P_i} \to \mathbb{R}$ is the *intrinsic cost function* which is convex and encodes the agent's personal preference between plans. So, if the agent was able to plan in isolation, it would obtain its optimal decision by finding the solution of the convex optimisation problem $\operatorname{argmin}_{p_i \in F_{P_i}} c_i(p_i)$. However, in a multi-agent environment, the agents affect each other. Therefore, in addition to its intrinsic cost, player P_i will attempt to minimize cost terms which arise from interactions with other agents; we will define these extra terms as soft constraints below and add them to P_i's objective. We assume that the intrinsic cost function and feasible region F_{P_i} are part of P_i's type. That is, they are private information and the agent may choose to truthfully reveal it, but may also choose to be silent or to lie. Therefore, our planning mechanism will be unable to rely on knowing each agent's type to achieve coordination.

We assume each feasible set F_{P_i} is a subset of some finite-dimensional $\mathbb{R}$-Hilbert space V_{P_i} with standard scalar product $\langle \cdot, \cdot \rangle_{V_{P_i}}$. We write $V = V_{P_1} \times \ldots \times V_{P_k}$ and $F_P = F_{P_1} \times \ldots \times F_{P_k}$ for the *overall* planning space and feasible region, and $p = (p_1^\intercal, \ldots, p_k^\intercal)^\intercal$ for the overall plan, respectively.

We model the coupling among players by a set of n soft linear interaction constraints. For ease of exposition, we will interpret them as "resource consumption" constraints. The soft constraints act as additional cost terms that discourage agents from preferring plans that jointly over-consume common resources.[3]

For each interaction constraint $j \in \{1, ..., n\}$ and player agent with index i, we assume that there is a convex function $\psi_{j,i} : V_{P_i} \to \mathbb{R}$ such that the consumption of resource j by plan $p_i \in V_{P_i}$ is $\psi_{j,i}(p_i)$. So, the total consumption of resource j amounts to $\psi_j(p) := \sum_i \psi_{j,i}(p_i)$. The j^{th} soft interaction constraint will penalise joint plans that imply a total consumption of resource j exceeding its system-wide supply level y_j. That is, we assume that there are continuous, convex penalty functions $\beta_j(\cdot)$ and scalars $y_j \in \mathbb{R}$ so that the overall cost due to consumption of resource j in plan p is $\beta_j(\psi_j(p) - y_j)$.

[3] Note that this does allow us to model resource production as well since production of m units of a resource can be modeled as the consumption of $-m$ units. We will henceforth focus our exposition on resource "consumption" only, tacitly assuming that it is understood that production can be modeled as the consumption of a negative amount.

For convenience, we define

$$\nu_j(p) = \psi_j(p) - y_j \tag{5}$$

which allows us to write the penalty cost pertaining to the j interaction constraint as $\beta_j(\nu_j(p))$.

If the j^{th} interaction constraint is the soft version of an inequality constraint $\nu_j(p) \leq 0$ then penalty function β_j should assume positive values for positive inputs (thereby penalising violations of the constraint) and be zero otherwise. For example, we could choose the hinge loss function $\beta_j : \nu \mapsto \max_{\lambda \in [0, M_j]} \lambda \nu$. (Note, this choice would correspond to the constraint penalty term in the Lagrangian dual of an optimisation problem.)

With these preliminary definitions, we define an *optimal overall coordination outcome* $p^* \in F_P$ to be the solution to the convex programme:

$$p^* \leftarrow \min_{p \in F_P} \omega(p) \tag{6}$$

where

$$\omega(p) = \sum_{i=1}^{k} c_i(p_i) + \sum_{j=1}^{n} \beta_j(\nu_j(p)) \tag{7}$$

is a convex function. Since this objective function is the sum of all of the individual agents' costs (plus the regularising terms that penalise outcomes that violate interaction-constraint violations), we will often refer to the optimal outcome p^* as the *socially optimal* or *efficient* solution and to $\omega(p)$.

The convex programme defines the gold standard of our coordination mechanism. To obtain this optimal outcome, in principle, we can attempt to obtain an optimal direct mechanism by evoking the revelation principle. That is, we let each agent P_i report its type (c_i, F_{P_i}) to a central authority which computes an efficient coordination outcome p^* by solving the centralised convex programme and informing the agents of their parts of their outcome. However, as explained above, this approach can often be undesirable for reasons of tractability (in terms of computation, memory or communication bandwidth) or due to a lack of trust or even knowledge on the part of the agents and the central authority.

Therefore, we propose an alternative route where the agents are re-incentivsed to learn to adjust their plans over the course of a repeated game-playing interaction. And, we will be able to provide (sub-) optimality guarantees on the ensuing joint outcomes as a function on the agent's overall learning performance (measured in terms of regret).

Put in mechanism design terms, we will design an indirect, dynamic, distributed mechanism that (at least approximately) implements the social choice function

$$\sigma : \big((c_1, F_{P_1}), \ldots, (c_k, F_{P_k})\big) \to \text{argmin}_{p \in F_P} \omega(p).$$

Furthermore, if the agents employ no-regret learning we provide guarantees describing how the approximation error vanishes with increasing length of the agents' interaction.

3 Our learning-based mechanism

In this section, by dualizing our soft resource constraints, we decouple the problem of finding a socially optimal plan. The result is a *saddle-point* or *minimax* problem whose variables are the original plan vector p along with new dual variables a, defined below. By associating the new variables a with additional agents, called *adversarial agents*, we arrive at a convex game with comparatively-sparse interactions. Based on this game, we introduce our proposed learning-based mechanism.

3.1 Introduction of adversarial agents

We define the *convex conjugate* or *dual* [BV04] of a function $\Gamma(x)$ to be $\Gamma^*(y) = \sup_{x \in \text{dom}\, \Gamma}[\langle x,y \rangle - \Gamma(x)]$. The conjugate function Γ^* is always closed and convex, and if Γ is closed and convex, then $\Gamma^{**} = \Gamma$ pointwise.

Write $F_{A_j} = \text{dom}\, \beta_j^*$. Since we have assumed that β_j is continuous and convex, we know that $\beta_j^{**} = \beta_j$ pointwise, that is, $\beta_j(\nu) = \sup_{a_j \in F_{A_j}}[a_j \nu - \beta_j^*(a_j)]$ for all $\nu \in \mathbb{R}$. Since β_j^* will become part of the objective function for the adversarial agents, and since many online learning algorithms require compact domains, we will assume that $F_{A_j} = [\underline{u}_j, \overline{u}_j]$ for some scalars $\underline{u}_j < \overline{u}_j$. (For example, in case the soft constraint is implemented by a hinge-loss, $\beta_j(x) = \max(0, Lx)$, we choose $\underline{u}_j = 0, \overline{u}_j = L$.) We will also assume that β_j^* is continuous on its domain.

We define $\Omega_{A_j} : V \times F_{A_j} \to \mathbb{R}$ as

$$\Omega_{A_j}(p, a_j) = a_j \nu_j(p) - \beta_j^*(a_j) \ . \tag{8}$$

And, writing $a = (a^1; \ldots; a^n) \in F_A = F_{A_1} \times \ldots \times F_{A_n}$, we define

$$\Omega : \begin{cases} F_P \times F_A \to \mathbb{R} \\ (p, a) \mapsto \sum_i c_i(p_i) + \sum_{j=1}^n \Omega_{A_j}(p, a_j) \end{cases} \tag{9}$$

(note the inclusion of the intrinsic costs $c_i(p_i)$). Because of the duality identity mentioned above, along with our assumption about dom β_j^*, we know that for all plans p,

$$\sup_{a_j \in F_{A_j}} \Omega_{A_j}(p, a_j) = \beta_j(\nu_j(p)) \tag{10}$$

and so

$$\omega(p) = \max_{a \in F_A} \Omega(p, a) \ , \forall p \in F_P. \tag{11}$$

Note that we have replaced $\sup$ by $\max$ in Eq. 11: since $\Omega(p, \cdot)$ is a closed concave function, it achieves its supremum on a compact domain such as F_A.

Remark 3.1. *Note, Eq. 11 establishes the connection between saddle-points of Ω and overall player plans that collectively minimize total cost in nature: If $(\tilde{p}, \tilde{a})$ is a saddle-point then we have*

$$\tilde{p} = \arg\min_{p \in F_P} \max_{a \in F_A} \Omega(p, a) = \arg\min_{p \in F_P} \omega(p). \tag{12}$$

(Cf. [Roc70]).

Now, we will create a new system that contains new *adversarial* agents that ensure the original *player* agents collectively incur Ω as their social cost at each stage of a repeated game-playing interaction. Here, we will ensure that the new agents are incentivised to adaptively incentivise the original player agents to avoid interaction constraint violations.

To this end, for the j^{th} interaction constraint ($j = 1, .., n$), we introduce an adversarial agent A_j that controls the parameter $a_j \in F_{A_j}$. To give all agents in this new system the desired incentives, we need to define transfers from player agents to adversarial agents (and vice versa) after each stage t of the game. The transfers will be defined such that the following holds:

1. Each player agent P_i ends up with a convex stage cost function $\Omega_{P_i(t)} : F_{P_i} \times F_A \to \mathbb{R}$ such that $\sum_i \Omega_{P_i(t)}(p_i, a) = \Omega(p, a), \forall p \in F_P, a \in F_A$. Note, given adversarial decision a, $\Omega_{P_i(t)}$ exclusively depends on the ith player's plan p_i.

2. Each adversarial agent A_j has a revenue proportional to $\Omega_{A_j}(p, a_j) + f_j(p)$ for all a_j. Here, $f_j(p)$ might summarize payments designed to compensate the players for any costs of their plans they might have incurred for the interaction before the mechanism was set up. For example, if penalty $\beta_j(\nu_j(p))$ is incurred by the players then $f_j(p)$ will equal $-\beta_j(p)$ (see below). Note that $f_j(p)$ does not depend on a_j, and so does not affect the choice of a_j once p is fixed. Therefore, for any player decision p, A_j is incentivised to maximise $\Omega_{A_j}(p, \cdot)$. Since this would in turn maximise the social cost Ω of the player agents, the adversarial agents are indeed the adversaries of the players in the game.

Obviously, the transfers have to depend on the costs in nature. That is, on the costs the agents incurred before the introduction of the mechanism.

We give two examples:

- Scenario 1: Before the introduction of the mechanims, each player P_i incurs an individual cost "in nature" $\omega_{P_i}(p) = c_i(p_i) + \beta_{ji}(p)$ where the β_{ji} encode agent P_i's share of the penalty $\beta_j(p)$ and $\sum_i \beta_{ji}(\cdot) = \beta_j(\cdot)$ (for details refer to [CG08b]). Then the social cost is just the sum of the individual costs in nature: $\omega(p) = \sum_i \omega_{P_i}(p)$. In this scenario, the transfers have to compensate P_i for its individual cost in nature. To this end, we ask P_i to make a net transfer to adversary A_j totalling $a_j\psi_{j,i}(p_i) - d_{ji(t)}r_j(a_j) - \beta_{ji}(p)$ where the *remainder function* $r_j(a_j) := a_j y_j + \beta_j^*(a_j)$ and the non-negative weights $d_{ji(t)}$ satisfy $\sum_i d_{ji(t)} = 1$ for each j (recall that y_j is the system tolerance for the j^{th} resource, and $\psi_{i,j}(p_i)$ is the total consumption of resource j by agent i under plan p_i). Thereby, the sum of all payments from all the players to A_j is $\sum_i a_j\psi_{j,i}(p_i) - d_{ji(t)}r_j(a_j) - \beta_{ji}(p) = \Omega_{A_j}(p, a_j) - \beta_j(\nu_j(p))$.

  Conversely, if we add up all costs of P_i (in nature and due to the transfer in the mechanism), agent P_i's cost in the mechanism is:

$$\begin{aligned}\Omega_{P_{i(t)}}(p_i, a) &= \omega_{P_i}(p) + \sum_{j=1}^{n} a_j\psi_{j,i}(p_i) - d_{ji(t)}\, r_j(a_j) - \beta_{ji}(p) \\ &= c_i(p_i) + \sum_{j=1}^{n}(a_j\psi_{j,i}(p_i) - d_{ji(t)}\, r_j(a_j)). \end{aligned} \tag{13}$$

  It is easy to verify that $\Omega(p, a) = \sum_i \Omega_{P_{i(t)}}(p_i, a)$ as desired.

  We can interpret the above payments as follows: A_j sets the per-unit price a_j for resource j. P_i pays A_j according to consumption, $a_j\psi_{j,i}(p_i)$, and, in return, is compensated for her share of the actual resource cost in nature, $\beta_{ji}(p)$. In addition, A_j pays a reimbursement fee of $d_{ji}r_j(a_j)$ to each player P_i for its share of the capacity of resource j. It is easy to show [CG08a] that the fee is always nonnegative. Furthermore, in several important scenarios, such as the market application we will present below, the remainder fee $r_j(a_j)$ equals $a_j y_j$.) Since the entire revenue for the agents A_j arises from payments by the player agents, we can think of A_j as opponents for the players—this is qualitatively true even though our game has many players and even though the player agent payoff functions contain terms that do not involve a.

- Scenario 2: There is no pre-mechanism player cost other than $c_i(p_i)$. In this case, the net transfers from P_i to A_j simply can be chosen to be $a_j\psi_{j,i}(p_i) - d_{ji(t)}r_j(a_j)$. These cause A_j to have $\Omega_{A_j}(p_i, a_j)$ as its revenue while ensuring $\Omega_{P_{i(t)}}(p_i, a) = c_i(p_i) + \sum_{j=1}^{n}(a_j\psi_{j,i}(p_i) - d_{ji(t)}\, r_j(a_j))$ as before.

In summary, we note several important properties:

- First, P_i's cost does not depend on any components of p other than p_i, and A_j's revenue does not depend on any components of a other than a_j. So, given an adversarial play a, each player agent P_i could plan by independently optimizing $\Omega_{P_i}(p_i, a)$. Similarly, given a plan p, each adversary could separately optimize $\Omega_{A_j}(p, a_j)$. So, the players' optimization problems are *decoupled* given a, and the adversaries' optimization problems are *decoupled* given p. This means that, in the game, the players play against the adversaries but not against each other.

- Second, if an adversarial agent A_j plays optimally, her revenue will maximise Ω_{A_j}, i.e. $a_j \in \text{argmax}_{a_j \in [0, u^j]} \Omega_{A_j}(p, a_j)$.

- Third, we have ensured that the total cost incurred by all player agents is at stage t for jointly playing p when the adversaries jointly play a is

$$\sum_i \Omega_{P_{i(t)}}(p_i, a) = \Omega(p, a) = \sum_i c_i(p) + \sum_j \Omega_{A_j}(p, a_j). \tag{14}$$

  If the adversaries each play optimally,

$$\Omega(p, a) = \sum_i c_i(p) + \sum_j \max_{a_j} \Omega_{A_j}(p, a_j) \overset{(10)}{=} \sum_i c_i(p) + \beta_j(p) = \omega(p).$$

So if the adversarial agents play optimally the total cost to the player agents is $\omega(p)$, just as it was in our original planning problem.

- Finally, since $\Omega(p, a)$ is a continuous saddle-function on a compact domain [Roc70], it must have a saddle-point $(\tilde{p}, \tilde{a})$. (By definition, a saddle-point of function $\Omega : F_P \times F_A \to \mathbb{R}$ is a point $(\tilde{p}, \tilde{a})$ such that $\Omega(p, \tilde{a}) \geq \Omega(\tilde{p}, \tilde{a}) \geq \Omega(\tilde{p}, a)$ for all $p \in F_P$ and $a \in F_A$.) By the decoupling arguments above, we must have that $\tilde{p}_i \in \arg\min_{p_i \in F_{P_i}} \Omega_{P_i}(p_i, \tilde{a})$ for each i, and that $\tilde{a}_j \in \arg\max_{a_j \in F_{A_j}} \Omega_{A_j}(\tilde{p}, a_j)$ for each j. (The latter is true since Ω and Ω_{A_j} differ only by terms that do not depend on a_j.) Therefore, $(\tilde{p}, \tilde{a})$ is a Nash-equilibrium of the game and the player part $\tilde{p}$ is a socially optimal solution of the original planning problem. That is, $\tilde{p} \in \text{argmin}_p \max_a \Omega(p, a) = \text{argmin}_p \omega(p)$.

3.2 Coordination as learning in a repeated game

We now consider our planning problem as a game among all of the agents P_i and A_j where Ω is the payoff function of the players. Via the consideration of saddle-points of Ω, we have just argued that there exists a Nash equilibrium $(\tilde{p}, \tilde{a})$ in pure strategies that simultaneously has the property that its player plan part $\tilde{p}$ must minimize $\omega(\tilde{p})$ and therefore, is socially optimal. (Note however, this does not mean that all Nash-equilibria of the game correspond to socially optimal solutions). To allow the agents to find such an equilibrium, we now cast the planning problem as learning in a repeated game. We will show that, if each agent employs a decision making algorithm that guarantees to incur sub-linear regret (e.g. a no-regret algorithm in the full information model), the agents as a whole will converge to a socially optimal plan, both in the sense that the average joint plan converges and in the sense that the average social cost converges. (This result, while similar to well-known results about convergence of no-regret algorithms to minimax equilibrium [FS96], does not follow from these older results, in part, because our game is neither constant-sum nor linear.)

When agents only receive partial information feedback during the game-playing interaction (e.g. if each agent is only informed about the cost of a plan rather than the entire cost function) to date only probabilistic confidence bounds on the regret are available. To accommodate such cases, we state our guarantees probabilistically for the most part.

Note that, from the individual agent's perspective, playing in the repeated game is an OCP, and so using a no-regret learner would be a reasonable choice; we explore the effect of this choice in more detail below in Sec. 4.

The repeated game playing protocol: The repeated game is played between the k players and the n adversarial agents over the course of multiple rounds we refer to as trials or stages. Based on their local histories of past cost feedback, in each stage t of the repeated game, each player P_i chooses a current pure strategy $p_{i(t)} \in F_{P_i}$, and simultaneously, each adversary A_j chooses a current resource price $a_{j(t)} \in F_{A_j}$. We write $p_{(t)} = (p_{1(t)}; ...; p_{k(t)})$ and $a_{(t)} = (a_{1(t)}; ...; a_{n(t)})$ for the joint actions of all players and adversaries, respectively.

After the players and adversaries have committed to $p_{(t)}$ and $a_{(t)}$, respectively, the transfers are made which result in observed cost and revenue feedback that can be added to the agents' information sets. The agents then update their decisions in light of the new information. Then, the system enters stage $t+1$, and the process repeats.

In the online setup, the stage plans are actually executed in each stage t and costs incurred immediately. By contrast, in the negotiation setup, the stage plans are mere proposals. Upon termination at time T, agent P_i is asked to compute and execute its $\bar{p}_{i,[T]} := \sum_{t \leq T} p_{i(t)}$

In the negotiation setup, the negotiation will have to terminate at some point. The termination criteria may vary. In the simplest case, a negotiation will end after a fixed number T of iterations. Otherwise, it might stop when all agents agree it should terminate (e.g. when all agents see that their average cost or plan has converged).

3.3 Full- versus partial- information feedback

The nature of the cost feedback depends on the question of whether the agents operate in the partial- or full-information model which will be outlined next:

3.3.1 Full-information feedback:

Here the player agents P_i get to observe the full cost *function* $\Omega_{P_i}(\cdot, a_{(t)})$. Similarly, a full-information model adversary A_j gets to observe his full reward function $\Omega_{A_j}(\cdot, p_{(t)})$.

One way of realising this would be as follows: Assuming its known to them, the players send all the information required to compute Ω_{A_j} to A_j: this includes the current resource consumptions $\psi_{j,i}(p_{i(t)})$. In return, each A_j sends their current price rate a_j to the players. In the online model, and if it is incurred in nature (see Scenario 1 above), P_i also observes $\beta_{ji}(p_{(t)})$ and sends it to A_j as well. The above information allows each P_i to compute its current *stage cost* function $\Omega_{P_i(t)}(\cdot) = \Omega_{P_i}(\cdot, a_{(t)})$ and its cost $\Omega_{P_i(t)}(p_{i(t)})$. It also allows each A_j to compute $\Omega_{A_j(t)}(\cdot) = \Omega_{A_j}(p_{(t)}, \cdot)$ as well as $\beta_{j(t)} = \beta_j(\nu_j(p_{(t)}))$, and thus, its total revenue. (In fact, A_j may avoid computing or storing $\beta_{j(t)}$ if desired, since it does not influence that term directly.) Next, we discuss how to implement the necessary communication efficiently.

Communication effort In this scenario, we have assumed that all player agents communicate their resource usages (and possibly their natural costs) to all adversarial agents, and all adversarial agents broadcast their prices to all player agents. Generally, this might involve $\mathcal{O}(nk)$ messages being sent per iteration.

However, if broadcasts are possible this effort can be substantially reduced. In this case, on every time step, each player sends one broadcast of size $\mathcal{O}(n)$ (her resource usages $\psi_{1,i}(p_{i(t)}), \ldots, \psi_{n,i}(p_{i(t)})$) and receives n messages of size $\mathcal{O}(1)$ (the resource prices), while each adversary sends one broadcast of size $\mathcal{O}(1)$ and receives k messages of size $\mathcal{O}(1)$, for a total of $n + k$ broadcasts per step, and a total incoming bandwidth of no more than $\mathcal{O}(\max\{n, k\})$.[4] Even under this simple assumption, the cost is somewhat better than a centralized planner, which would have to receive k much larger messages describing each player's detailed optimization problem, and send k much larger messages describing each player's optimal plan.

However, by exploiting *locality*, we can reduce bandwidth even further: in many problems we can guarantee *a priori* that player P_i will never use resource j, and in this case, we never need to transmit A_j's price a_j to P_i. Similarly, if P_i decides not to use resource j on a given trial, we never need to transmit $\psi_{j,i}(p_{i(t)})$ to A_j. (To take full advantage of locality, we must also set the weights d_{ji} so that players do not receive payments from adversaries they would otherwise not need to talk to.) So, by using targeted multicasts instead of broadcasts, we can confine each player's messages to a small area of the network; in this case, no single node or link will see even $\mathcal{O}(k + n)$ traffic. We can sometimes reduce bandwidth even further by combining messages as they flow through the network: for example, two resource consumption messages destined for A_j may be combined by adding their reported consumption values.

Finally, any implementation needs to make sure that the agents cannot gain by circumventing the mechanism: e.g., no player should find out another's plan before committing to her own.[5]

3.3.2 Partial-information model

To realise the previous setting, one has to assume that all agents are aware of the structure of their individual cost functions and the structure of the constraints. In order to compute $\psi_{j,i}(p_{i(t)})$ from its plan $p_{i(t)}$ and to evaluate its current cost function $\Omega_{P_i(t)}$ player agent P_i has to be aware of the $\psi_{j,i}$ for all possible inputs. This may not always always realistic either for computational reasons or lack of knowledge about the environment.

For example, consider a situation where the agents are threads running on a parallel computer. The plans are which processors to employ to perform computation. The cost is run-time. The soft interaction constraints quantify not only delays due to over-subscriptions of the nodes (e.g. cores and GPUs) but also delays due to congestion in the communication pathways (e.g. the bus) connecting the nodes. The latter might will typically vary from architecture to architecture and therefore, it cannot be assumed that the agents are aware of this a priori. Another example scenario will be given in our experimental section. Therein, we consider a situation where the agents'

[4] Technically, the agents could multicast rather than broadcast, so that, e.g., one player would never see another player's messages, but in practice one would not expect this optimization to save much.

[5] One of the undesirable effects resulting if we would permit agents to wait until all other agents have sent their plans before choosing an own plan is that we could run into a deadlock. In general, we could enforce a commitment to a certain plan prior to learning about the other agents' current choices by employing cryptographic methods such as commitment schemes [Blu81, Eve82, Nao91].

plans might affect the production output of a plant and the coordination mechanism desires on average to limit the resulting emission of green house gases which are taxed. Here, the relationship between production output control decisions and emissions might be unknown. So the agents might have to rely on measurements of emissions in response to their decision (and the resulting penalty tax amount) as feedback.

In such cases, it may be more realistic to assume that all agents get to observe a sample of their stage cost functions. For instance, player agent P_i may only receive a " bill" of $\Omega_{P_i(t)}(p_{i(t)})$ for its proposed stage plan $p_{i(t)}$ (bandit feedback). In this case, the agent might choose to employ a sublinear-regret incurring bandit algorithm to update its stage proposal plans (e.g. they might use OPSGD). If all agents choose to do so, then we will still be able to give (usually probabilistic) bounds on the coordination success of the outcome dynamics.

Communication effort Changing to the partial-information setting does not necessarily change then number of messages that need to be sent in the system. However, depending on the precise setup, it might affect the size of the message being sent. For example, consider a setting where adversary A_j is aware of $\psi_{j,i}$ but player P_i oblivious to it. To get to observe cost feedback, P_i might sent her full plan $p_{i,(t)}$ to A_j (who then responds by computing and sending the the "bill" $\Omega_{P_i(t)}(p_{i(t)})$ back to player P_i). Therefore, the message size has increased from $\mathcal{O}(1)$ (for the player sending $\psi_{j,i}(p_{i(t)})$ to $\mathcal{O}(\dim F_{P_i})$. Note, if the player plan involves at least one component for each interaction constraint then $\dim F_{P_i} \in \Omega(n)$.

4 Theoretical Analysis

Up to this point, we have described the mechanism both for the online and for the negotiation setup. The question remains what the effects of the interaction are and how the decision-making algorithms as well as the length of the interaction impact the quality of the overall solution in terms of social utility. In this section, we investigate how the regret properties of the decision algorithms the agents employ can bound the social cost of the overall plans evolves, as functions of both T and δ, resulting from the repeated game-playing interaction.

4.1 Game between two synthesized agents

For analysis, it will help to construe our set-up as a fictitious game between a *synthesized player agent*, P, and a *synthesized adversarial agent*, A. When each component agent P_i plays $p_{i(t)}$ and each component agent A_j plays $a_{j(t)}$, then we imagine P to play $p_{(t)}$ and A to play $a_{(t)}$. Accordingly, we understand P to incur cost $\Omega(p_{(t)}, a_{(t)})$, and A to have revenue proportional to $\Omega_A(p_{(t)}, a_{(t)})$ in round t. These synthesized agents are merely theoretical notions serving to simplify our reasoning; in practice there would never be a single agent controlling all players.

Using these synthesized agents, we will prove two results. First, immediately below, we show that if there exist regret bounds for the individual component agents, we can derive regret bounds for the synthesized agents. Second, we show that sub-linear regret bounds for the synthesised can be converted into guarantees on the proximity to equilibrium of the game for large values of T.

Depending on whether the component agents employ no-regret algorithms with deterministic or high-probability regret bounds, all our results will be deterministic or probabilistic statements. To formalise this we introduce the following shorthand for statements about an algorithm $\mathcal{A}$:

A(Δ, t, δ) The algorithm $\mathcal{A}$ satisfies the regret bound $R_{\mathcal{A}}(T) \leq \Delta(T, \delta)$, with probability $1 - \delta$.

Note that the above can be either a *deterministic* or *high-probability* assertion about the algorithm $\mathcal{A}$, depending on whether δ is equal to 1, or lies in $(0, 1)$, respectively.

4.1.1 From component to synthesized agents

If all the agents employ algorithms satisfying regret bounds of the form **A**(Δ, t, δ), we can exploit the decoupled property of the cost functions and use simple union bounds to obtain upper bounds on the regrets of the synthesized player and adversarial agents. For the player agent we have:

Lemma 4.1. *Let $\delta \in [0,1)$, $T \in \mathbb{N}$ and $a_{(1)}, \dots, a_{(T)} \in F_A$. Suppose each player agent P_i employs an algorithm satisfying the statement **A(**Δ_{P_i}, T, δ**)**. Then the synthesized player agent P achieves a regret bound*

$$\mathcal{R}_P(T) = \sum_{t=1}^{T} \Omega(p_{(t)}, a_{(t)}) - \min_p \sum_{t=1}^{T} \Omega(p, a_{(t)}) \le \sum_{i=1}^{k} \Delta_{P_i}(T, \delta) =: \Delta_P(T),$$

with probability at least $1 - k\delta$.

Proof. Recall that the regret for player P_i is

$$\mathcal{R}_{P_i}(T) = \textstyle\sum_{t=1}^{T} \Omega_{P_i(t)}(p_{i(t)}, a_{(t)}) - \min_{p_i,} \sum_{t=1}^{T} \Omega_{P_i(t)}(p_i, a_{(t)}).$$

Owing to the decoupling effect of the adversary we have $\Omega(p, a_{(t)}) = \sum_{i=1}^{k} \Omega_{P_i(t)}(p_i, a_{(t)})$. So, we can expand $R_P(T)$ as

$$\begin{aligned}
\mathcal{R}_P(T) &= \sum_{t=1}^{T} \Omega(p_{(t)}, a_{(t)}) - \min_{p \in F_P} \sum_{t=1}^{T} \Omega(p, a_{(t)}) \\
&= \sum_{t=1}^{T} \sum_{i=1}^{k} \Omega_{P_i(t)}(p_{i(t)}, a_{(t)}) - \min_{p \in F_P} \sum_{t=1}^{T} \sum_{i=1}^{k} \Omega_{P_i(t)}(p_i, a_{(t)}) \\
&= \sum_{i=1}^{k} \sum_{t=1}^{T} \Omega_{P_i(t)}(p_{i(t)}, a_{(t)}) - \sum_{i=1}^{k} \min_{p_i \in F_{P_i}} \sum_{t=1}^{T} \Omega_{P_i(t)}(p_i, a_{(t)}) \\
&= \sum_{i=1}^{k} \big(\sum_{t=1}^{T} \Omega_{P_i(t)}(p_{i(t)}, a_{(t)}) - \min_{p_i \in F_{P_i}} \sum_{t=1}^{T} \Omega_{P_i(t)}(p_i, a_{(t)}) \big) = \sum_{i=1}^{k} \mathcal{R}_{P_i}(T).
\end{aligned}$$

Using a union bound and the assumption that, for all players P_i, $\Pr[R_{P_i}(T) \le \Delta_{P_i}(T, \delta)] \ge 1 - \delta$, we conclude that:

$$\begin{aligned}
\Pr[\mathcal{R}_P(T) > \Delta_P(T, \delta)] &= \Pr\left[\sum_{i=1}^{k} \mathcal{R}_{P_i}(T) > \sum_{i=1}^{k} \Delta_{P_i}(T, \delta) \right] \\
&\le \Pr\left[\sum_{i=1}^{k} \mathcal{R}_{P_i}(T) > \sum_{i=1}^{k} \Delta_{P_i}(T, \delta) \right] \\
&\le \Pr\left[\exists i \in \{1, ..., k\} : \mathcal{R}_{P_i}(T) > \Delta_{P_i}(T, \delta) \right] \\
&\le \sum_{i=1}^{k} \Pr[\mathcal{R}_{P_i}(T) > \Delta_{P_i}(T, \delta)] \le k\delta.
\end{aligned}$$

□

Analogously, for the adversarial agent A we have the following lemma. The proof is very similar, and is therefore omitted.

Lemma 4.2. *Let $\delta \in [0,1)$ and $T \in \mathbb{N}$. Suppose each player agent A_i employs an algorithm satisfying the statement **A(**Δ_{A_i}, T, δ**)**. Then the synthesized player agent A achieves a regret bound*

$$\mathcal{R}_A(T) \le \sum_{i=1}^{k} \Delta_{A_i}(T, \delta) =: \Delta_A(T),$$

with probability at least $1 - k\delta$.

In the following we typically desire all agents to incur sub-linear regret, so that the system will converge to the desired social optimum. We therefore merge the two lemmas above into a single guarantee about overall regret:

Lemma 4.3. *With the notation and under the assumptions of Lemmas 4.1 and Lem. 4.2 we have:*

$$\Pr\Big[\mathcal{R}_P(T) \le \Delta_P(T,\delta) \wedge \mathcal{R}_A(T) \le \Delta_A(T,\delta)\Big] \ge 1-(n+k)\delta.$$

Proof.

$$\begin{aligned}
&\Pr\Big[\mathcal{R}_P(T) \le \Delta_P(T,\delta), \mathcal{R}_A(T) \le \Delta_A(T,\delta)\Big]\\
&= 1-\Pr\Big[\mathcal{R}_P(T) > \Delta_P(T,\delta) \vee \mathcal{R}_A(T) > \Delta_A(T,\delta)\Big]\\
&\ge 1-\Pr\Big[\mathcal{R}_P(T) > \Delta_P(T,\delta)\Big] - \Pr\Big[\mathcal{R}_A(T) > \Delta_A(T,\delta)\Big] \ge 1-k\delta-n\delta
\end{aligned}$$

where the first inequality is an application of the union bound and the second inequality an application of Lem. 4.1 and Lem. 4.2. □

4.1.2 From synthesized player regret to global social cost bounds

Now we investigate the behaviour and cost performance of the averaged synthesized strategies

$$\bar{p}_{[T]} := \frac{1}{T}\sum_{t=1}^{T} p_{(t)} \text{ and } \bar{a}_{[T]} := \frac{1}{T}\sum_{t=1}^{T} a_{(t)},$$

as well as the averaged costs $\frac{1}{T}\sum_{t=1}^{T}\Omega(p_{(t)}, a_{(t)})$. (Recall that the negotiation version of our mechanism outputs the averaged averaged costs.)

First we need to show that $\max_a \min_p \Omega(p,a) = \min_p \max_a \Omega(p,a)$. To do so, we note that It is standard knowledge that for any real-valued function κ on a nonempty product set $X \times Y$, we have $\sup_{x\in X} \inf_{y\in Y} \kappa(x,y) \le \inf_{y\in Y} \sup_{x\in X} \kappa(x,y)$ [Roc70]. For our case, this fact translates to $\max_{a\in F_A} \min_{p\in F_P} \Omega(p,a) \le \min_{p\in F_P} \max_{a\in F_A} \Omega(p,a)$, since we work with compact sets and continuous functions.

We now have the means to show that the minmax and the maxmin value of Ω coincide. Such a result was first proven in von Neumann's well-known minimax theorem ([vN28]) for the case of two-player, zero-sum matrix games.

Theorem 4.4 (Modification of Von Neumann's minimax theorem [vN28])**.** *We have* $\max_a \min_p \Omega(p,a) = \min_p \max_a \Omega(p,a)$.

Proof. By construction of Ω we have: $\forall a \in F_A : \Omega(\cdot, a) : F_P \to \mathbb{R}$ convex and $\forall p \in F_P : \Omega(p,\cdot) : F_A \to \mathbb{R}$ concave.

Recall that there exists an algorithm satisfying the statement **A(**$\Delta_{A_i}, T, 0$**)** for all values of T. Therefore, we can learn to play the game employing this algorithm.

Now, for any T, we have

$$\begin{aligned}
\min_p \max_a \Omega(p,a) &\leq \max_a \Omega(\bar{p}_{[T]}, a) = \max_a \Omega\left(\frac{1}{T}\sum_{t=1}^{T} p_{(t)}, a\right) &(15)\\
&\leq \max_a \frac{1}{T}\sum_{t=1}^{T} \Omega(p_{(t)}, a) &(16)\\
&\leq \frac{1}{T}\sum_{t=1}^{T} \Omega(p_{(t)}, a_{(t)}) + \frac{\Delta_A(T,\delta)}{T} &(17)\\
&\leq \min_p \frac{1}{T}\sum_{t=1}^{T} \Omega(p, a_{(t)}) + \frac{\Delta_P(T,\delta)}{T} + \frac{\Delta_A(T,\delta)}{T} &(18)\\
&\leq \min_p \Omega(p, \bar{a}_{[T]}) + \frac{\Delta_P(T,\delta)}{T} + \frac{\Delta_A(T,\delta)}{T} &(19)\\
&\leq \max_a \min_p \Omega(p,a) + \frac{\Delta_P(T,\delta)}{T} + \frac{\Delta_A(T,\delta)}{T}, &(20)
\end{aligned}$$

where (16) follows since $\Omega(\cdot, a)$ is convex $\forall a$, (17) and (18) follow from Lemma 4.2 and Lemma 4.1, respectively, and (19) follows since $\Omega(p, \cdot)$ is concave $\forall p$.

Hence, letting $T \to \infty$ we can conclude $\min_p \max_a \Omega(p,a) \leq \max_a \min_p \Omega(p,a)$. This concludes the result. □

The following theorem establishes high-probability bounds both on the sub-optimality of the average social cost incurred in online mode as well as on the sub-optimality of the negotiation outcome $(\bar{p}_{[T]}, \bar{a}_{[T]})$ as a function of the negotiation duration T.

Theorem 4.5. *Let $\delta \in [0,1)$ and $T \in \mathbb{N}$ and $v = \min_{p \in F_P} \max_{a \in F_A} \Omega(p,a)$. Suppose each player agent P_i employs an algorithm satisfying the statement **A(**Δ_{P_i}, T, δ**)**, and each adversarial agent A_j employs an algorithm satisfying the statement **A(**Δ_{A_j}, T, δ**)**.*

Then, setting $\epsilon_P := \frac{\Delta_P(T,\delta)}{T}$ and $\epsilon_A := \frac{\Delta_A(T,\delta)}{T}$, we have:

1. $\Pr\Big(\frac{1}{T}\sum_{t=1}^{T} \Omega(p_{(t)}, a_{(t)}) \in [v - \epsilon_A, v + \epsilon_P]\Big) \geq 1 - (n+k)\delta$,
2. $\Pr\Big(\omega(\bar{p}_{[T]}) \in [v, v + \epsilon_P + \epsilon_A]\Big) \geq 1 - (n+k)\delta$, *and*
3. $\Pr\Big(\Omega(\bar{p}_{[T]}, \bar{a}_{[T]}) \in [v, v + 2(\epsilon_P + \epsilon_A)]\Big) \geq 1 - (n+k)\delta$.

From this result it is clear that if regret guarantees of the form **A(**Δ, T, δ**)** on the algorithms used by the component agents that are sub-linear in T, then meaningful worst-case bounds on the global costs in both the online and offline settings can be given.

Proof of Thm. 4.5. By Lem. 4.3, with probability of at least $1 - (n+k)\delta$, $\Delta_P(T,\delta) \leq \mathcal{R}_P(T)$ and $\Delta_A(T,\delta) \leq \mathcal{R}_A(T)$. Now, assuming that these inequalities hold, then so does the derivation (15) to (20).

1. From (15) to (20) we can extract that:

$$v \leq \frac{1}{T}\sum_{t=1}^{T} \Omega(p_{(t)}, a_{(t)}) + \frac{\Delta_A(T,\delta)}{T} \leq v + \frac{\Delta_P(T,\delta)}{T} + \frac{\Delta_A(T,\delta)}{T},$$

   which proves the first statement of the theorem.

2. From (15) to (20) we can extract that:

$$v \leq \max_a \Omega(\bar{p}_{[T]}, a) = \omega(\bar{p}_{[T]}) \leq v + \frac{\Delta_P(T,\delta)}{T} + \frac{\Delta_A(T,\delta)}{T},$$

which proves the second statement of the theorem.

3. Furthermore, we can now derive that

$$\begin{aligned} v &\leq \min_p \Omega(p, \bar{a}_{[T]}) + \frac{\Delta_P(T,\delta)}{T} + \frac{\Delta_A(T,\delta)}{T} \\ &\leq \Omega(\bar{p}_{[T]}, \bar{a}_{[T]}) + \frac{\Delta_P(T,\delta)}{T} + \frac{\Delta_A(T,\delta)}{T} \\ &\leq \max_a \Omega(\bar{p}_{[T]}, a) + \frac{\Delta_P(T,\delta)}{T} + \frac{\Delta_A(T,\delta)}{T} \\ &\leq v + 2\frac{\Delta_P(T,\delta)}{T} + 2\frac{\Delta_A(T,\delta)}{T} \end{aligned}$$

which proves the third statement of the theorem.

□

If the agents use algorithms that require the time horizon T as a parameter, then, provided the satisfy its conditions, Thm. 4.5 already provides all the information we need about the cost performance of the mechanism to decide how long to run for in order to achieve a desired level of sub-optimality with a given probability.

4.1.3 Convergence guarantees

The previous theorem gave a (probabilistic) bound on the sub-optimality of the mechanism as a function of the number of agents and the planning length quantified by the number of negotiation trials T. We found that if all agents employ no-regret-algorithms during planning that, for *a single negotiation*, the planning outcomes approximate optimal decisions within an error that vanishes in the limit of longer planning duration with a confidence defined by the δ parameter.

In what is to follow, we collect some further notions of *asymptotic consistency* of the mechanism under various assumptions on the known regret bounds of the algorithms the agents employ.

(A) (Strongly sub-linear regret bounds) There exists a function $f : \mathbb{N} \to \mathbb{R}$ for which $\sum_{T=1}^{\infty} f(T) < \infty$, and, for all $T \in \mathbb{N}$, all player agents P_i and adversarial agents A_j use algorithms satisfying **A(**$\Delta_{P_i}, T, f(T)$**)** and **A(**$\Delta_{A_j}, T, f(T)$**)**, respectively, where $\Delta_{P_i}(T, f(T)), \Delta_{P_i}(T, f(T)) = o(T)$.

Normally, when high probability regret bounds are given for learning algorithms in OCP problems, the dependency on δ scales like $log(1/\delta)$. Hence, taking $f(T) = 1/T^2$ in assumption **(A)** would be perfectly reasonable. Of course, a special case is when we have deterministic regret bounds, and we can restate this in the following stronger assumption:

(A)' (Deterministic regret bounds) For all $T \in \mathbb{N}$, all player agents P_i and adversarial agents A_j use algorithms satisfying **A(**$\Delta_{P_i}, T, 0$**)** and **A(**$\Delta_{A_j}, T, 0$**)**, respectively, where $\Delta_{P_i}(T, 0), \Delta_{P_i}(T, 0) = o(T)$.

Corollary 4.6. *If (A)' holds, then, we have, with probability* 1*:*

1. *The average payoffs converge to the minimax value,* $\frac{1}{T}\sum_{t=1}^{T} \Omega(p_{(t)}, a_{(t)}) \overset{T\to\infty}{\longrightarrow} v$*, with a worst-case rate given by* $\max\{\frac{\Delta_A(T,\delta)}{T}, \frac{\Delta_P(T,\delta)}{T}\}$.
2. *The cost in nature* $\omega(\bar{p}_{[T]})$ *of the players' part of the negotiation outcome converges to the optimal cost in the limit of long negotiations,* $\omega(\bar{p}_{[T]}) \overset{T\to\infty}{\to} v$*, with a worst-case rate of at most* $\frac{\Delta_P(T,\delta)}{T} + \frac{\Delta_A(T,\delta)}{T}$.

3. *The payoff of the negotiation outcome converges to the value of the game in the limit of a large number of negotiation rounds T : $\Omega(\bar{p}_{[T]}, \bar{a}_{[T]}) \overset{T\to\infty}{\to} v$ with a worst-case rate of at most $2\frac{\Delta_P(T,\delta)}{T} + 2\frac{\Delta_A(T,\delta)}{T}$.*

*If only **(A)** holds, then the statements of convergence above still hold, but the worst case rate statements can be violated at most finitely often, with probability* 1.

Proof. The statements under the assumption **(A)'** follow trivially from Thm. 4.5.

Now suppose that only the assumption **(A)** holds. Let $\mathcal{E}_{T,f(T)}$ be the event that $\mathcal{R}_{P_i} < \Delta_{P_i}(T, f(T))$ and $\mathcal{R}_{A_j} < \Delta_{A_j}(T, f(T))$ for all player agents P_i and adversarial agents A_j. Then, since $\sum_{T=1}^{\infty} f(T) < \infty$ we have from the Borell-Cantelli Lemmas that

$$\Pr\Big[\mathcal{E} := \{\mathcal{E}^c_{T,f(T)} \text{ occurs for only finitely many } T\}\Big] = 1.$$

The result now follows from applying Thm. 4.5 on the event $\mathcal{E}$. ☐

As a consequence of this corollary, if all player and adversarial agents satisfy **(A)**, we can guarantee convergence of $(\bar{p}_{[T]}, \bar{a}_{[T]})$ to a set $K_P \times K_A$ of saddle-points of Ω (where $K_P \subset F_P, K_A \subset F_A$). Specifically, while the sequences $\bar{p}_{[T]}$ and $\bar{a}_{[T]}$ may not converge, the distance of $\bar{p}_{[T]}$ from K_P and the distance of $\bar{a}_{[T]}$ from K_A approach zero, and every cluster point of the sequence $(\bar{p}_{[T]}, \bar{a}_{[T]})$ is in $K_P \times K_A$. Due to the convexity and compactness of the feasible sets, each average strategy and each cluster point is feasible.

Theorem 4.7. *Under the assumption (A), the averaged synthesized strategies $(\bar{p}_{[T]}, \bar{a}_{[T]})$ converge to a (bounded) subset $K_P \times K_A$ of saddle-points of the player cost function $\Omega : F_P \times F_A \to \mathbb{R}$, with probability 1.*

Proof. Let the event $\mathcal{E}$ be as defined in Cor. 4.6, and assume that it holds.

Recall that , $F_P \subset V, F_A \subset \mathbb{R}^n$ denote the compact, convex feasible sets of the synthesized player P and adversary A, respectively. Let the set of all saddle-points of Ω in $F_P \times F_A$ be denoted by E:

$$E := \big\{(p, a) \in F_P \times F_A | \forall \tilde{p} \in F_P \forall \tilde{a} \in F_A : \Omega(p, \tilde{a}) \leq \Omega(p, a) \leq \Omega(\tilde{p}, a)\big\}$$

(cf. [Roc70]) which correspond to the minimax-equilibria of the game. Let

$$K_P := \big\{p \in V | \forall e > 0 \text{ and } t \in \mathbb{N},\ \exists T > t : \big\|p - \bar{p}_{[T]}\big\| < e\big\}$$
$$\text{and } K_A := \big\{a \in \mathbb{R}^n | \forall e > 0 \text{ and } t \in \mathbb{N},\ \exists T > t : \big\|a - \bar{a}_{[T]}\big\| < e\big\},$$

be the set of all cluster points of the sequence $(\bar{p}_{[T]})_{T\in\mathbb{N}}$, and $(\bar{a}_{[T]})_{T\in\mathbb{N}}$, respectively. So, we want to prove that $K_P \times K_A \subset E$, and, to do so, we show that $\forall \bar{p} \in K_P, \bar{a} \in K_A$,

$$\max_a \Omega(\bar{p}, a) \leq \Omega(\bar{p}, \bar{a}) \leq \min_p \Omega(p, \bar{a}).$$

We introduce the well-defined functions: $\psi : F_A \to \mathbb{R}, a \mapsto \min_p \Omega(p, a)$ and $\phi : F_P \to \mathbb{R}, p \mapsto \max_a \Omega(p, a)$ and note, $\forall a, p : \psi(a) \leq \Omega(p, a) \leq \phi(p)$. Now, by Theorem 4.5, setting $\epsilon_T = (\Delta_P(T, f(T)) + \Delta_A(T, f(T)))/T$,

$$v \leq \max_a \Omega(\bar{p}_{[T]}, a) \leq v + \epsilon_T, \text{ and } v \leq \min_p \Omega(p, \bar{a}_{[T]}) + \epsilon_T \leq v + \epsilon_T$$

Hence, the following limits exist and are as follows:

$$\lim_{T\to\infty} \phi(\bar{p}_{[T]}) = \lim_{T\to\infty} \max_a \Omega(\bar{p}_{[T]}, a) = \max_a \min_p \Omega(p, a) = v \tag{21}$$

$$v = \max_a \min_p \Omega(p, a) = \lim_{T\to\infty} \min_p \Omega(p, \bar{a}_{[T]}) = \lim_{T\to\infty} \psi(\bar{a}_{[T]}). \tag{22}$$

Let $\tilde{p} \in K_P, \tilde{a} \in K_A$ be arbitrary choices. By definition of the cluster sets there exists a subsequence $(s_T)_{T=1}^{\infty}$ of $(\bar{p}_{[T]})_{T=1}^{\infty}$ and a subsequence $(u_T)_{T=1}^{\infty}$ of $(\bar{a}_{[T]})_{T=1}^{\infty}$, such that

$$s_T \to \tilde{p} \text{ and } u_T \to \tilde{a} \text{ as } T \to \infty.$$

We know Ω, ϕ, ψ are continuous (cf. Sec. 3 and Appendix B, respectively). Hence,

$$v = \lim_{T\to\infty} \psi(\bar{a}_T) = \lim_{T\to\infty} \psi(u_T) = \psi(\tilde{a}).$$

Analogously, we obtain: $v = \phi(\tilde{p})$. Hence, $v = \phi(\tilde{p}) \geq \Omega(\tilde{p}, a)$, $\forall a$ and thus,

$$v \geq \Omega(\tilde{p}, \tilde{a}) \geq \psi(\tilde{a}) = v.$$

With the help of (21) and (22), we can now deduce that $(\tilde{p}, \tilde{a})$ is a saddle point:

$$\max_a \Omega(\tilde{p}, a) = \phi(\tilde{p}) = v = \Omega(\tilde{p}, \tilde{a}) = v = \psi(\tilde{a}) = \min_p \Omega(p, \tilde{a}).$$

Since $\tilde{p}, \tilde{a}$ were arbitrary choices we conclude that $K_P \times K_A \subset E$. □

Since Ω is continuous on its domain, Thm. 4.7 lets us conclude that the outcome of negotiation is a plan which approximately minimises total player cost in nature: if we choose T sufficiently large, then $(\bar{p}_{[T]}, \bar{a}_{[T]})$ must be close to a saddle-point, and so the costs must be close to the costs of the of a saddle-point. (To determine how large we need to choose T, we can look at the regret bounds of the learning algorithms of the individual agents – this will be done below.) As expressed in Rem. 3.1, the player part $\tilde{p}$ of a saddle-point incurs minimal total player cost in nature and, since the adversarial agents learn to set prices as nature would, $\tilde{p}$ is socially optimal (w.r.t. to the player agents) with respect to both nature and the mechanism.

4.2 Convergence to Nash-equilibrium

Above we have given guarantees of convergence to a saddle-point of Ω which was proven to also correspond to the socially optimal coordination outcome. In the negotiation approach the (approximate) saddle-point is the result of the interaction. However, when working with selfish, strategic agents, we want to know whether a selfish agent has an incentive to unilaterally deviate from its part of the negotiation outcome. The following theorems show that all saddle-points are Nash-equilibria and therefore, the answer is, at least approximately, *no*: in the limit of large T, the negotiation outcomes $\bar{p}_{[T]}, \bar{a}_{[T]}$ converge to a subset of Nash equilibria (under the conditions we provided to ensure convergence to the set of saddle-points). So, by continuity, $(\bar{p}_{[T]}, \bar{a}_{[T]})$ is an approximate Nash equilibrium for sufficiently large T—that is, each agent has a vanishing incentive to deviate unilaterally.

Recall that F_{P_i} denotes the feasible set of player agent P_i $(i \in \{1, ..., k\})$, F_{P_j} denotes the feasible set of player agent P_j $(j \in \{1, ..., n\})$, ad K_P and K_A denote the sets of cluster points of the averaged synthesized player and adversarial strategies respectively.

Theorem 4.8. *Let $(\tilde{p}, \tilde{a})$ be a saddle-point of $\Omega(p, a)$ with respect to minimizing over p and maximizing over a. For all $i \in \{1, \ldots, k\}$ and stages t we have:*

$$\Omega_{P_i(t)}(\tilde{p}_i, \tilde{a}) = \min_{p'_i \in F_{P_i}} \Omega_{P_i}(p'_i, \tilde{a})$$

Proof. Let $\tilde{p} \in K_P, \tilde{a} \in K_A$ and $t \in \mathbb{N}$. We know from Thm. 4.7 that $(\tilde{p}, \tilde{a})$ is a saddle-point of Ω with respect to minimization over F_P and maximization over F_A. Hence, for all $p' \in F_P$: $\Omega(\tilde{p}, \tilde{a}) \leq \Omega(p', \tilde{a})$. Since $\Omega(p, a) = \sum_m \Omega_{P_m(t)}(p_m, a), \forall p, a, t$, we have in particular that, for all $p'_i \in F_{P_i}$,

$$\Omega_{P_i(t)}(p'_i, \tilde{a}) + \sum_{m \neq i} \Omega_{P_m(t)}(\tilde{p}_m, \tilde{a}) \geq \Omega(\tilde{p}, \tilde{a}) = \Omega_{P_i(t)}(\tilde{p}_i, \tilde{a}) + \sum_{m \neq i} \Omega_{P_m(t)}(\tilde{p}_m, \tilde{a}).$$

Thus, $\Omega_{P_i(t)}(p'_i, \tilde{a}) \geq \Omega_{P_i(t)}(\tilde{p}_i, \tilde{a})$, for all $p'_i \in F_{P_i}$. □

Theorem 4.9. *Let $(\tilde{p}, \tilde{a})$ be a saddle-point of $\Omega(p, a)$ with respect to minimizing over p and maximizing over a. We have, for all $j \in \{1, ..., n\}$,*

$$\Omega_{A_j}(\tilde{p}, \tilde{a}_j) = \max_{a_j \in F_{A_j}} \Omega_{A_j}(\tilde{p}, a_j).$$

Proof. Notice that

$$\sum_{j=1}^{n} \Omega_{A_j}(\tilde{p}, \tilde{a}_j) = \max_{a \in F_A} \sum_{j=1}^{n} \Omega_{A_j}(\tilde{p}, \tilde{a}_j) = \sum_{j=1}^{n} \max_{a^j \in F_{A_j}} \Omega_{A_j}(\tilde{p}, a_j)$$

where the last equality follows since the adversarial revenue functions are independent.

So we have $\sum_{j=1}^{n} (\max_{a^j \in F_{A_j}} \{\Omega_{A_j}(\tilde{p}, a^j)\} - \Omega_{A_j}(\tilde{p}, \tilde{a}_j)) = 0$. On the other hand, for all j, we have that $\max_{a^j \in F_{A_j}} \{\Omega_{A_j}(\tilde{p}, a^j)\} - \Omega_{A_j}(\tilde{p}, \tilde{a}_j) \geq 0$. Hence, for all j, $\max_{a^j \in F_{A_j}} \{\Omega_{A_j}(\tilde{p}, a^j)\} - \Omega_{A_j}(\tilde{p}, \tilde{a}_j) = 0$. □

Theorem 4.9 allows us to conclude that the individual part of an adversarial agent's negotiation outcome is a best-response action:

Corollary 4.10. *Let $(\tilde{p}, \tilde{a})$ be a saddle-point of $\Omega(p, a)$ with respect to minimizing over p and maximizing over a. We have, for all $j \in \{1, ..., n\}$,*

$$\Omega_{A_j}(\tilde{p}, a_j) \leq \Omega_{A_j}(\tilde{p}, \tilde{a}_j).$$

4.3 Runtime analysis

In this subsection, we will discuss computational aspects of our mechanism. Previously, we have derived bounds on the (sub-) optimality of the coordination performance. These bounds depend on regret bounds of the agents' decision algorithms and thereby, on the number of iterations T the mechanism is run.

It will therefore be interesting to consider the question how many of those iterations the mechanism has to be run in order to achieve a desired approximation level to the optimal outcome. Of course the answer will depend on the regret bounds of the algorithms utilised by the agents. For the two particular algorithms discussed above, in Sec. 4.3.1, we will provide such an analysis that gives a bound T_ϵ on the number of iterations to achieve a coordination outcome whose social cost is within an ϵ-interval of the minimal attainable social cost.

Since we are interested in multi-agent coordination. Another important question is that of computational scalability of our approach in the number of player agents k to be to coordinated (and indeed the number of interaction constraints n which we assume to be independent of k for ease of exposition). Note, that since the computation in each stage can be performed in parallel by all the agents and since the agents' problems are decoupled, T_ϵ will dominate the run-time complexity of the coordination as a function of k.

As we will see below, T_ϵ will depend on k and n. So, we will write $T_\epsilon(k, n)$. Furthermore, $T_\epsilon(k, n)$ depends on the worst regret bound of any of the agents in the system. To make the exposition concrete, it will therefore be convenient to assume the agents all use a particular no-regret learning algorithm in each stage of the game. For convenience, let us assume that each agent uses Greedy Projection (i.e. OPGD). As will be shown in Cor. 4.13 below in case all agents employ OPGD, $T_\epsilon(k, n) \in \mathcal{O}((k + n)^2)$ rounds. Now, per iteration, each agent has to run OPGD. Since the agents can perform all computation in parallel the effort for performing an OPGD update step is independent of the number of player agents k (but might scale linearly with n). Since the agents can do their calculations in parallel in each round, the runtime per iteration will be in $\mathcal{O}(n)$ (independent of k). Put together, the total run-time of our mechanism scales as $\mathcal{O}(n(n + k)^2)$.

Of course, we can simulate the mechanism on a single core and use it as a centralised convex optimisation method. In the absence of leveraging parallelism, this would mean that the resulting method would scale as $\mathcal{O}(kn(n + k)^2)$.

Remark 4.11. Note, that in this computation the quadratic dependence on k stems from the $\sqrt{T}$-dependence of the regret bound of OPGD. For particular classes convex cost functions this bound can be substantially improved [HAK07]. As a result, for coordination problems that involve cost functions belonging to such sub-classes, we will be able to derive much better run-time bounds.

Remark 4.12. Of course, part of the motivation for running a distributed method such as ours lies in the expectation of better scalability compared to centralised optimisation of the objective. For instance, if we wanted to solve the centralised optimisation problem (cf. Eq. 6) directly, we would have to perform approximate convex optimisation of a dk-dimensional problem. Without exploitation of the composite structure of the feasible set, in general and up to this point, convex optimisation seems to exhibit a worst-case complexity that depends on factors (such as Lipschitz constants, dimensionality, feasible set diameters) that, put together, result in overall computational effort that grows worse than cubically with k.

It remains to derive the bound T_ϵ. For the no-regret algorithms considered in this paper, this will be done next.

4.3.1 Bounds on the number of coordination iterations as a function of the desired level of sub-optimality – results for specific no-regret algorithms

Recall that Thm. 4.5 provides a means to convert high-probability regret bounds on the algorithms employed by the individual agents into high-probability bounds on the social cost incurred. In this section, we provide a corollary to this result showing for how long one must run a negotiation phase in order to achieve a certain desired level of accuracy when employing the specific no-regret algorithms introduced in Section 2.

Corollary 4.13 (Number of negotiations with OPGD)**.** *Suppose that the individual players' feasible sets and loss functions satisfy the conditions of Thm. 2.2, and suppose that all agents employ the Online Projected Gradient Descent (OPGD) algorithm given in Algorithm 1. Let L denote the maximum Lipschitz constant of any of the player and adversarial agents' payoff functions and let D denote the maximum Euclidean diameter of all feasible sets of the agents, i.e. $D := \max\{diamF_{A_j}, diamF_{P_i} | i = 1, ..., k, j = 1, ..., n\}$.*

Then, for any $\epsilon > 0$, choosing $T \geq (k+n)^2 \left(D + 2L^2\right)^2 / \left(4\epsilon^2\right)$ we guarantee that

$$\frac{1}{T}\sum_{t=1}^{T} \Omega(p_{(t)}, a_{(t)}) \in [v - \epsilon, v + \epsilon],\ \omega(\bar{p}_{[T]}) \in [v, v + 2\epsilon],$$
$$\text{and } \Omega(\bar{p}_{[T]}, \bar{a}_{[T]}) \in [v, v + 4\epsilon].$$

Proof. Recall from Thm. 2.2 that the regret of

$$\mathcal{R}_P(T) + \mathcal{R}_A(T) \leq \Delta(T) = (k+n)\left(\frac{D}{2} + L^2\right)\sqrt{T},$$

where R is the maximum diameter of all the agents' feasible sets, and L is an upper-bound on the gradient of the agents' cost functions. By Thm. 4.5, setting T such that

$$\epsilon \geq \frac{\Delta(T)}{T} \iff \epsilon \geq (k+n)\left(\frac{D}{2} + L^2\right)\frac{1}{T^{\frac{1}{2}}} \iff T \geq \left((k+n)\frac{D + 2L^2}{2\epsilon}\right)^2$$

we have proved the result. $\square$

We can provide a similar result for the bandit-information setting, where the agents receive only their costs as feedback.

Corollary 4.14 (Number of negotiations with OPSGD)**.** *Suppose that the individual players' feasible sets and loss functions satisfy the conditions of Thm. 2.3, and suppose that all agents employ the Online Projected Stochastic Gradient Descent (OPSGD) algorithm given in Algorithm 2. With other notation being as before, let C denote the maximum value of any of the agents' absolute costs. Then, for any $\epsilon > 0$, $\delta \in (0, 1/(k+n))$, choosing*

$$T \geq \max\left\{\left((n+k)4\left(RdC\left(3 + \frac{R}{r}\right)L\right)^{\frac{1}{2}}\right)^5 \epsilon^{-5}, \delta^{-20}\right\}$$

guarantees that, with probability $1-(k+n)\delta$*, we have*

$$\frac{1}{T}\sum_{t=1}^{T}\Omega(p_{(t)},a_{(t)})\in[v-\epsilon,v+\epsilon],\ \omega(\bar{p}_{[T]})\in[v,v+2\epsilon],$$
$$\textit{and } \Omega(\bar{p}_{[T]},\bar{a}_{[T]})\in[v,v+4\epsilon].$$

Proof. Recall from Thm. 2.2 that the regret of

$$\mathcal{R}_P(T)+\mathcal{R}_A(T)\leq\Delta(T,T^{-\frac{1}{20}})=(n+k)4\left(RdC\left(3+\frac{R}{r}\right)L\right)^{\frac{1}{2}}T^{\frac{4}{5}}.$$

So, by Thm. 4.5, if we setting T such that

$$\epsilon\geq\Delta(T,T^{-\frac{1}{20}})/T\iff\epsilon\geq(n+k)4\left(RdC\left(3+\frac{R}{r}\right)L\right)^{\frac{1}{2}}T^{-\frac{1}{5}}$$
$$\iff T\geq\left((n+k)4\left(RdC\left(3+\frac{R}{r}\right)L\right)^{\frac{1}{2}}\right)^{5}\epsilon^{-5}$$

we have proved the result. □

5 Experiments

Our approach is suitable for scenarios where an optimal solution could be found by solving a convex programme. Therefore, a diverse range of applications are conceivable. For illustration purposes, we choose three simulated coordination problems. First, we illustrate our method in the context of a emission taxation application. The second problem setting is a simple market of goods where our adversaries negotiate the prices. The final setting concerns (selfish) network routing where the agents plan paths of flow through a graph with interaction constraints on total amount of flow that is allowed to traverse each node.

5.1 Learning tax rates for emission reduction

In an ever growing global economy, the increased production of pollutants such as green house gas (GHG) emissions has long been recognised as a substantial ecological threat. In response, the global community has engaged in various governmental measures to reduce pollutant output in their economies in order to meet nationally or internationally agreed targets.

The governmental instruments for achieving the targets are diverse and range from taxation and market-based approaches to legislatively imposed caps on net emission levels. In carbon pricing alone the number of instruments expanded by 90% between 2012 and 2015, reaching a total value of just under 50 billion USD [ea15]. Two of the most widely accepted instruments are *taxation* on the one hand and *cap and trade* on the other. At this point we can only give a brief introduction to both approaches. For a more detailed introduction of the various mechanisms and a discussion of their merits and shortcomings, we refer the reader to [Gou13, Cen11, Keo09, CFHP10].

In cap and trade, the government that decides to regulate a particular industry under their jurisdiction decides on a fixed cap on the total volume of a pollutant (e.g. carbon) that the industry is jointly allowed to produce. A government institution then distributes emission certificates granting the right to produce a set amount of the pollutant within a given time period. The total number of certificates issued amount to emission allowances that equal the imposed cap. By law, the industry participants are not permitted to exceed the level of emissions granted by the certificates they hold. However, the participants are allowed to sell on certificates they do not need. The idea behind this approach is that this incentivises participants to make reductions in pollutant emissions in proportion to anticipated economic viability. That is, companies that can more easily implement emission savings will do so as they can expect to then sell on the certificates they hold to other participants for who it would be more costly to achieve the equivalent reductions. While cap and trade is an approach that is widely implemented, many open questions remain. For example, there are many different models of how the initial distribution of certificates should be conducted (e.g. by equal distribution, by auctions, fixed sale prices) and it is still under investigation [HS11, CFHP10] under which conditions and to what extent the various approaches indeed achieve an efficient (that is socially optimal) allocation, especially when agents act sub-optimally. Furthermore, it is not well understood how the rigid cap levels affect the industry as whole and whether less rigid methods such as taxation would not offer greater flexibility and economic stability [Gou13].

On the other hand, in a tax regime, a fixed tax rate is imposed on the production of each unit of the targeted pollutants. Some of the identified downsides of the tax approach appear to be (i) that it tends to be unclear a priori of how the choice of tax rate impacts the reduction in the pollutant relative to the target and, (ii) that it seems to be an open question in how far the tax rate introduces market inefficiencies that negatively affect the economy to an unnecessarily large extent.

In what is to follow, we apply our mechanism to the pollutant reduction task. Here, in a scheme we call *cap and tax*, we interpret the adversaries as tax authorities whose actions are setting tax rates on the overproduction of pollutants beyond a set cap. Due to our theoretical guarantees we can be confident that the tax agents can learn how to adapt these tax rates in order to serve an efficient allocation respecting the caps at least approximately (in the negotiation setup) or at least guarantee that the production caps are met on average in the long run (when the mechanism is employed in the online setup). Therefore, it addresses many of the open issues prevalent in the most common current approaches.

5.1.1 Our mechanism for cap and tax

We will now connect our mechanism to the cap and tax application domain. The player agents are stake-holders in an industry deemed to be affected by the pollutant cap. For example, an agent might be the owner of a power plant or a car factory. Player P_i's plan p_i is a decision that affects the production output, which in turn affects the agent's utility u_i (coinciding with the negative of its cost $c_i(p_i)$). As a byproduct, implementing p_i causes the emission of $\psi_{ji}(p_i)$ units of pollutant of variety j. The industry-wide production of each pollutant j is ideally to be capped by y_j. However, to allow for greater flexibility the government is content with imposing soft constraints. That is, it incentivises agents to adhere to the constraint via tax signals that are to be set according to our mechanism. To this end, for each pollutant j it installs a tax department (adversarial agent) A_j granted agency for setting the tax rate $a^j \in [0, \bar{u}_j]$ for some choice $\bar{u}_j$. Since the pollutant emission previously was an externality, the setting follows Scenario 2 of Sec. 3.1 and A_j's revenue is $\Omega_{A_j}(p, a_j) = a_j(\sum_{i=1}^{k} \psi_{ji}(p_i) - y_j)$. (This corresponds to a penalty function $\beta_j(p) = \max(0, \bar{u}_j \sum_{i=1}^{k} \psi_{ji}(p_i) - y_j)$ in the definition of the coordination objective.) Following the repeated game approach, the production levels of the pollutants (e.g. the $\psi_{ji}(p_i)$) are reported in regular time intervals (the stages of the game) after which the tax rates a^j are updated with a no-regret algorithm. The pertaining tax bill after stage t player P_i incurs is $\sum_{j=1}^{n}(a_j\psi_{j,i}(p_i) - d_{ji(t)}\, r_j(a_j))$ (the net transfers discussed above). Here, $r_j(a_j) = a_j y_j$ and, for simplicity we choose $d_{ji(t)} = \frac{1}{k}$.

As an illustration, we simulated a simple scenario with two player agents. Here, the industry comprises two players. Their cost functions are $c_1(p_1) = -p_{1,1}$ and $c_2(p_2) = -2p_{2,2}$ where $p_{i,k}$ denotes the k^{th} component of decision vector p_i. The decisions could be decisions on how to operate a plant that generates revenue (negative cost) but also pollutants as a byproduct. For simplicity, we assume their decisions directly correspond to the emission of pollutants. That is, $\psi_{j,i}(p_i) = p_{i,j}$. The first player agent was at liberty to choose actions independently, i.e. $F_{P_1} = [0, 1]^2$. So, in order gain maximum utility 1, the first player would desire to choose $p_1 = (1, x)^\top$ for any $x \in [0, 1]$. However, the second player's production process emits the two regulated pollutants in equal amounts. Therefore, his feasible set is $F_{P_2} = \{p_2 \in [0, 1]^2 \,|\, p_{2,1} = p_{2,2}\}$. Therefore, his maximum utility generating action is the single vector $(1, 1)^\top$. Given a global pollutant cap level of $y_1 = 1$ and that both players would desire to produce 1 unit of the first pollutant, coordination is required.

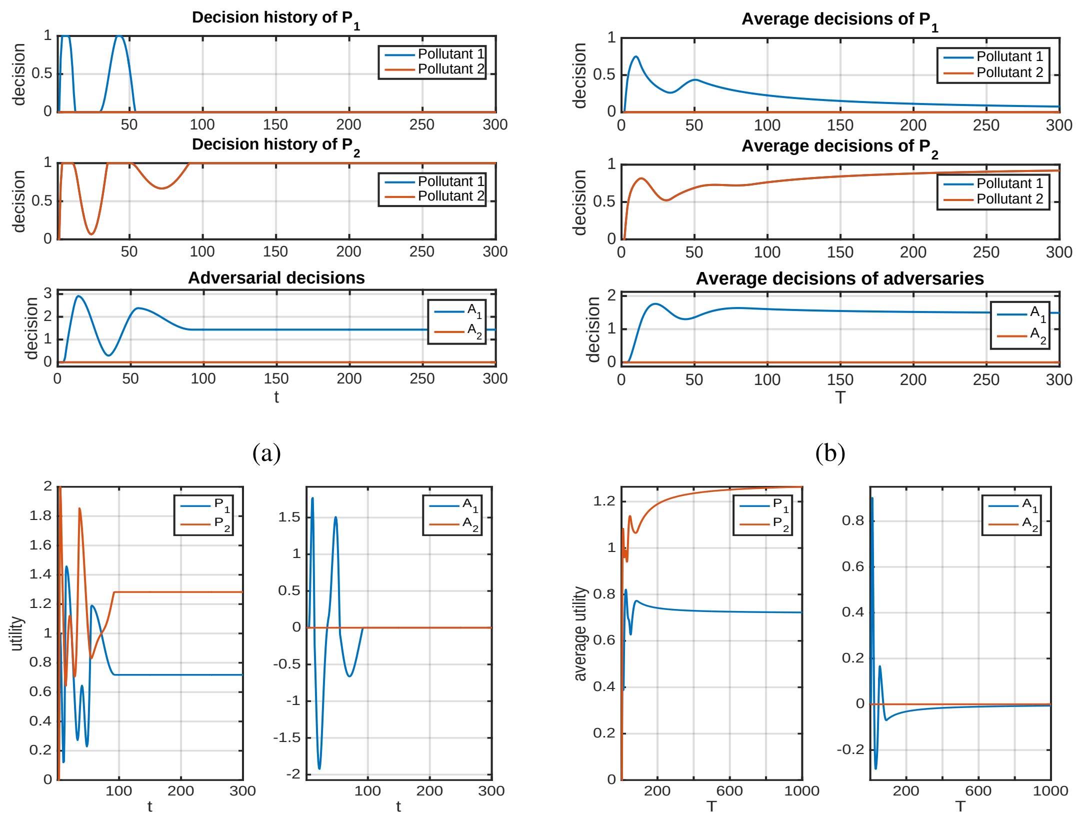
To this end, we instituted two adversaries with feasible sets $F_{A_1} = F_{A_2} = [0, 50]$.

With these parameters, the ideal global coordination objective (cf. Eq. 6) becomes

$$\min_{p \in F_P} \omega(p_1, p_2) = \min_{p_1 \in F_1, p_2 \in F_2} -p_{1,1} - 2\,p_{2,2} + 50 \max\{0, p_{1,j} + p_{2,j} - 1\}$$

which defines the desired optimal social choice function to be approximately implemented by our mechanism.

From the statement, we can see that the optimal coordination outcome p^* approximately is $p_1^* = (0,0)^\top, p_2 = (1,1)^\top$. That is, a situation where the first player agent seizes to produce anything in order to allow the second player to produce as much as he pleases to gain maximum utility. In this optimal solution, the social utility approximately amounts to 2.



(a) (b)

(c) (d)

Figure 1: The simulation of our mechanism in the cap and tax scenario. (a): Decisions of all the the agents for the different stages $t = 1, \ldots, 300$. (b) Average decisions as a function of negotiation length T. (c) Stage utilities for each agent. (d) Average utilities for each agent for horizon lengths $T = 1, \ldots, 1000$.

To achieve coordination, we simulated our mechanism with these parameters. To ensure no-regret during the repeated game, we let agents update their decisions with the Greedy Projection algorithm (Alg. 1). The resulting behaviour is depicted in Fig. 1. From the plots, we can see that the system behaves just as predicted by our theory. In particular, with increasing interaction length length T, the average player plans $\bar{p}_{[T]}$ (Subplot (b)) converge to an approximation of the optimal joint decision p^* and the average social utilities (Subplot (d)) $-\frac{1}{T}\sum_{t=1}^{T} \Omega(p_{(t)}, a_{(t)})$ converge to the negative value of the game, i.e. to $-v = -\min_p \max_a \Omega(p,a) = -\omega(p^*) = 2$, which coincides the social utility of the optimal joint outcome. Note, that P_1 still makes revenue (utility) at later stages of the game, even though he does not produce anything. This is due to the reimbursement $d_{1,1} r_1(a_1) = d_{1,1} y_1 a_1 = \frac{a_1}{2}$ he receives from the tax agent A_1. We can think of these payments as subsidies that aide a socially optimal outcome.

Furthermore, we can see that the average joint outcome $\bar{p}_{[T]}, \bar{a}_{[T]}$ is indeed an approximate Nash equilibrium for sufficiently large T: Consider $T = 300$; where $\bar{p}_{[T]} \approx p^* = (0,1)^\top$. Obviously, P_2 achieved its optimum and so has no incentive to deviate. So, we now consider agent P_1. With a tax rate of $\bar{a}_{[T],1} > 1$, the first player agent would only lose by increasing its output of pollutant 1 to say $\epsilon > 1$ units: while he would gain 1ϵ in intrinsic utility, P_1 would be taxed at a rate of $a_1\epsilon$ where $a_1 > 1$. Therefore, P_1 is better off not increasing its pollutant output. On the other end, since the players do not violate the soft constraints (i.e. since the sum of outputs of pollutant one and the sum of outputs of pollutant 2 each do not exceed cap level 1) the adversaries have no incentive to alter their tax rates as this would not generate additional tax revenue.

We note, that in order to be able to employ Greedy Projection, the player agents require full-information feedback (which in this case are the gradients of the cost functions with respect to the player's decisions). As such, P_i needs to be informed about $\nabla_{p_i}\psi_{ji}(p_i)$. However, in realty this functional relationship between a production decision and the emission of a pollutant may not be known. Instead the agent might make a decision and obtain measurement of the resulting emission. This would mean that the agent would have to base its decision on partial information. Hence, to study systems with this bandit feedback, simulations based on bandit algorithms might be more appropriate. In the next subsection, we will study the system behaviour resulting from using such bandit algorithms in the context of a simple market scenario.

5.2 A simple market scenario

Above we have given an example of a mechanism where the adversaries regulate emission caps via adaptive taxation. Other scenarios of interest are market situations where coordination is to be achieved via price negotiation between sellers and buyers of goods. For instance, in the cap and trade approach the goods might be emission allowances that are traded among the agents. The market is set up in the hope that the allowances will be distributed in a socially optimal manner.

Motivated by these applications, we consider a simple market containing two infinitely divisible goods.

Initially, one unit of each good is held by a seller agent $S_1 := P_2$. Its plan $p_2 \in F_{P_2} = [0,1]^2$ is a two-dimensional vector whose qth component encodes how much the agent is willing to sell of the qth good. The seller has a private utility $u_{2,j} \in \mathbb{R}$ for possessing each unit of good j. So, with cost coinciding negative utility, its private cost function is $c_2(p_2) = -\langle u_2, p_2\rangle$ where $u_2 = (u_{1,2}, u_{2,2})^\top$.

The market is also inhabited by two potential buyer agents, $B_1 := P_1$ and $B_2 := P_3$ whose plans are two-dimensional vectors whose components indicate how much of each good the agents intend to buy.

Having an internal valuation of $u_{j,i}$ per unit of good j acquired, each buyer agent P_i ($i \in \{1,3\}$) has an intrinsic cost function $c_i(p_i) = \langle -u_i, p_i\rangle$ for assessing plan p_i. In addition to constraining the amount of each resource to be bought to be less than 2 units but non-negative, seller agent P_2 also wishes to only sell the two goods in equal quantities. That is, the feasible sets are $F_1 = F_3 = \{x \in \mathbb{R}^2_{\geq 0} | x_1 \leq 2, x_2 \leq 2\}$ and $F_2 = F_1 \cap \{x \in \mathbb{R}^2 | x_1 = x_2\}$.

5.2.1 Our indirect negotiation mechanism

Instead of negotiating prices directly, we employ our coordination mechanism. to find a price for the goods that spawn planning outcomes that correspond to an efficient (re-) allocation of the goods and to ensure a match of supply and demand. That is, we desire an overall outcome $p \in F_1 \times F_2 \times F_3$ such that $\omega(p) = c_1(p_1) + c_2(p_2) + c_3(p_3) + \beta_1(\nu_1(p)) + \beta_2(\nu_2(p))$ is minimal. Here $\beta_j(\nu_j(p)) = \max_{\pi \in [-\bar{\pi}, \bar{\pi}]} \pi\big(p_1 + p_3 - p_2\big)$ is a soft-constraint penalty rate attempting to discourage supply-demand mismatches, i.e. outcomes violating the condition $p_2 = p_1 + p_3$. This means the capacity y_j will be set to zero (or, viewed from a different angle, the seller sets the capacity dynamically to foster its own interest). As we will see, π will correspond to a price for the contested goods that we will have negotiated by the adversarial agents.

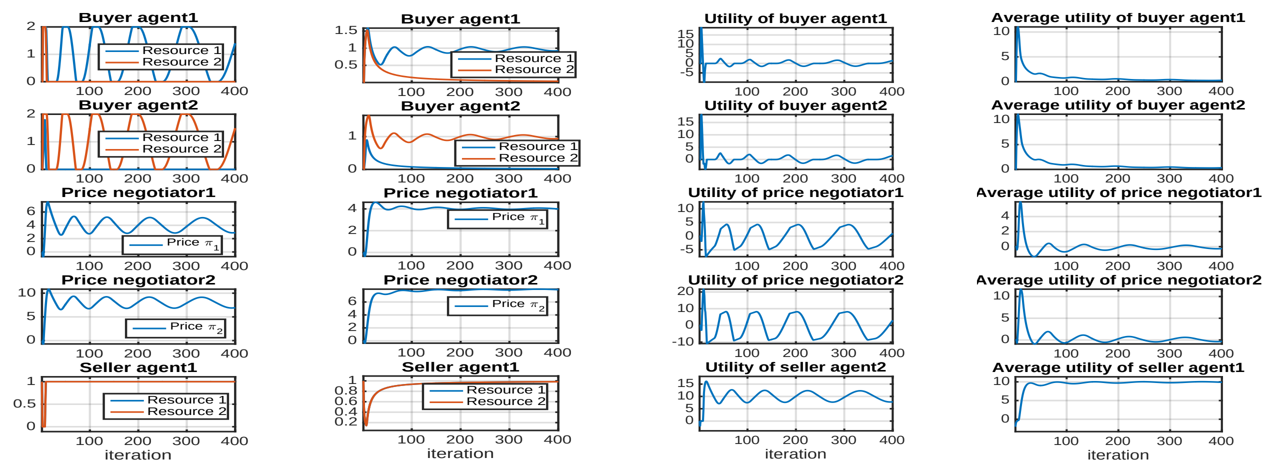
To this end, we introduce two adversarial agents A_1 and A_2 to negotiate per-unit prices π_1 and π_2 for goods 1 and 2, respectively (i.e. adversary j has action $a_j = \pi_j \in F_{A_j} := [0, 10]$).

For each $j \in 1, 2$, adversary A_j acts as a middle-man between the seller and the buyers of resource j. It chooses a price π_j per unit of resouce j it gives to the seller. So the sellers cost function is $\Omega_2(p_2) = c_2(p_2) - \sum_{j=1}^{2} \pi_j p_{2,j}$ where $\pi_j p_{2,j}$ is the profit the adversary A_j hands over to the seller P_2 for the $p_{2,j}$ units of good j sold.

The adversary then sells on the good to the buyers at the given price. Therefore, buyer P_i ($i \in \{1,3\}$) has a cost of $\Omega_{P_i} = c_i(p_i) + \sum_{j=1}^{2} \pi_j p_{i,j}$.

Being a middle-man the adversary A_j makes a revenue of $\Omega_{A_j}(\pi_j, p) = \pi_j\big(p_1 + p_3 - p_2\big)$ where $\pi_j\big(p_1+p_3-p_2\big)$ is the profit due to buying from seller P_2 and selling on to the buyers at price π_j. Note that when A_j plays optimally he plays $a_j = \text{argmax}_{\pi_j} \Omega(\pi_j, p) = \arg\max_{\pi_j \in [-\bar{\pi}, \bar{\pi}]} \big(p_1 + p_3 - p_2\big)\pi_j$ which is just the soft -constraint penalty for supply-demand mismatch. To ensure all plans are feasible in each round, we might consider a setting where the adversary is able to trade additional quantities of the good j with an external entity at a price $\bar{\pi}$. For instance, assume the plans of the buyers ask for too much of the resource j relative to the amount sold by the seller. That is, the discrepancy $d := p_1 + p_3 - p_2$ is positive. In that case, we can imagine that the adversarial agent might have the option of making good on his promise by purchasing the missing amount d from the external entity at price $\bar{\pi}$. So, $\max\{\big(p_1 + p_3 - p_2\big)\pi_j | \pi_j \in [-\bar{\pi}, \bar{\pi}]\}$ is just the expense for doing so. (Conversely, if the adversary bought too much, we can imagine that the external authority has to be paid for accepting the excess goods).

In this case, adversary A_j has the revenue $\Omega_{A_j}(\pi_j, p) = \pi_j\big(p_1 + p_3 - p_2\big) - \max\{\big(p_1 + p_3 - p_2\big)\pi_j | \pi_j \in [-\bar{\pi}, \bar{\pi}]\}$. This renders the setting consistent with Scenario 1, outlined in Sec. 3.



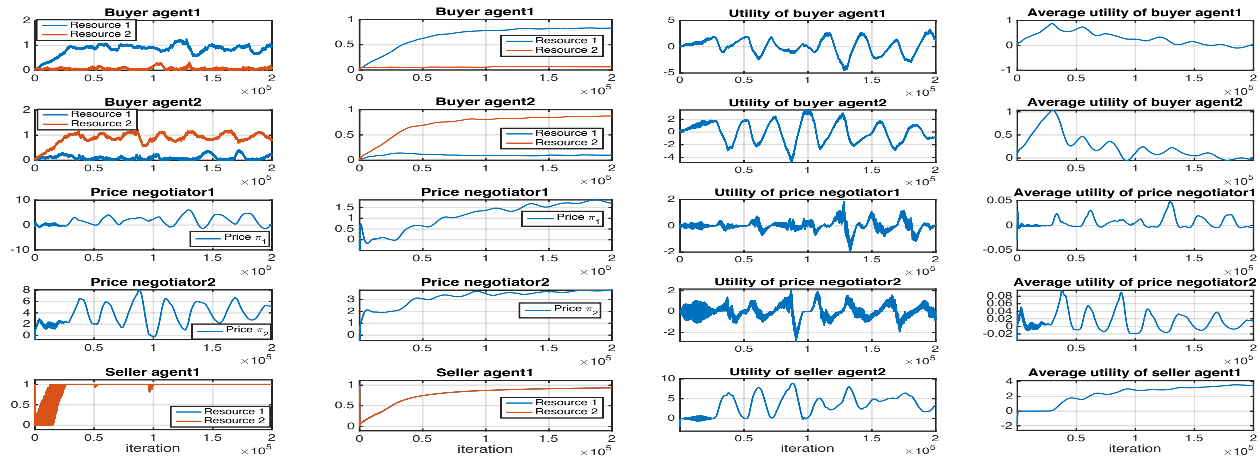
(a) (b) (c) (d)

Figure 2: The simulation of the negotiation in our first market scenario. (a): Decisions of all the the agents for the different stages (iterations) $t = 1, \ldots, 500$. (b): Average decisions as a function of negotiation length T. (c): Stage utilities. (d): Average utilities within the mechanism. All agents employed Greedy Projection for decision making. Adversarial agent A_j is called 'price negotiator j'.

An example simulation of this market, where all the agents update their decisions with Greedy Projection, is depicted in Fig. 2. For the simulation, the buyer utility vectors were $u_1 = (4,4)^\top, u_3 = (0.5, 8)^\top$. That is, in an optimal allocation, we expect agent buyer 1 to receive the full amount of 1 unit of resource 1 and the second buyer agent (P_3) to receive all of the second resource. On the other hand, the seller's utility for keeping the resource were 1. As before, the outcomes are consistent with our theorems and seem to converge to the optimal allocation. Furthermore, the sum of all utilities seems to converge to the maximal attainable social utility (or minimal social cost). Note, the system happens to favour the seller in terms of utility. However, in future work, we could investigate how to alter the overall coordination objective to facilitate fairness in addition to social optimality.

To investigate the partial-information setting we have repeated the simulation with buyer agents utilising the OPSGD bandit algorithm (Alg. 2) in place of Greedy Projection. This time, the buyer utility vectors were $u_1 = (2,2)^\top, u_3 = (0.5, 4)^\top$. The simulation results are depicted in Fig. 3.

While the system does eventually converge to an optimal solution it takes substantially longer than in the full-information setting. In the light of our analysis of Sec. 4.3 and, in particular, Cor. 4.14, this might not come as a great surprise. To render the bandit setting more viable from a computational perspective, future work would have to consider alternative bandit approaches that offer regret bounds more advantageous for rapid coordination.



(a) (b) (c) (d)

Figure 3: The simulation of the negotiation in our first market scenario but where the buer agents employ bandit algorithms to update their decisions. (a): Decisions of all the the agents for the different stages (iterations) $t = 1, \ldots, 500$. (b): Average decisions as a function of negotiation length T. (c): Stage utilities. (d): Average utilities within the mechanism.

5.3 Network routing

As a third example scenario, we conducted experiments on a small multi-agent min-cost routing domain.[6] We model our network by a finite, directed graph with edges E (physical links) and vertices V (routers). Each edge $e \in E$ has a finite capacity $\gamma(e)$, as well as a fixed intrinsic cost c^e for each unit of traffic routed through e. We assume that bandwidth is infinitely divisible.

Players are indexed by a source vertex s and a destination vertex r. Player P_{sr} wants to send an amount of flow d_{sr} from s to r. P_{sr}'s individual plan is a vector $f_{sr} = (f_{sr}^e)_{e \in E}$, where $f_{sr}^e \in [0, U]$ is the amount of traffic that P_{sr} routes through edge e. (U is an upper bound, chosen ahead of time to be larger than the largest expected flow.) Feasible plans are those that satisfy *flow conservation*, i.e., the incoming traffic to each vertex must balance the outgoing traffic. We also excluded plans which route flow in circles.

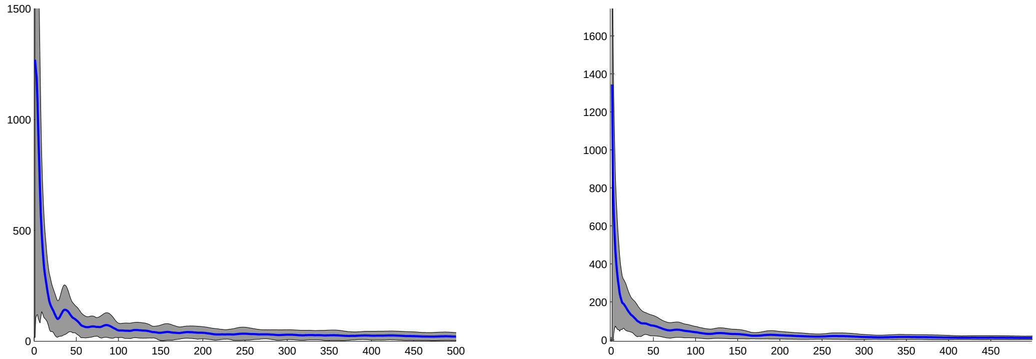
If the total usage of an edge e exceeds its capacity, all agents experience an increased cost for using e. This extra cost could correspond to delays, or to surcharges from a network provider. We set the global penalty for edge e to be a hinge loss function $\beta_e(\nu) = \max\{0, u_e \nu\}$, so the total cost of e increases linearly with the amount of overuse. For convenience we also modeled each player's demand d_{sr} as a soft constraint $\beta_{sr}(\nu) = \max\{0, u_{sr}\nu\}$, although doing so is not necessary to achieve a distributed mechanism.

Applying our problem transformation led to new, total player cost
$\Omega(f, a) = \sum_{s,r} \sum_{e \in E} c^e f_{sr}^e + \sum_{e \in E} \Omega_{A_e}(a_{\text{cap}}^e, f) + \sum_{s,r} \Omega_{A_{sr}}(a_d^{sr}, f)$ in the mechanism. Here, for each edge e, we introduced an adversarial agent A_e, who controls the cost for capacity violations at e by setting the price a_{cap}^e. And, as a slight extension to our general description in Section 3, we introduced additional adversarial agents to implement the soft constraints on demand; agent A_{sr} chooses a price a_d^{sr} for failing to meet demand on route sr.

With this setup, we ran more than 2800 simulations, for the most part on random problem instances, but also for manually-designed problems on graphs of sizes varying between 2 and 16 nodes. In each instance there were between 1 and 32 player agents. For no-regret learning, we used the Greedy Projection algorithm [Zin03] with $(\frac{1}{\sqrt{t}})_{t \in \mathbb{N}}$ as the sequence of learning rates.

A simple example of an averaged player plan after a number of iterations is depicted in Figure 5(b). In this experiment, we had a 6-node network and three players $P_{2,3}$, $P_{1,4}$, and $P_{4,6}$ with demands 30, 70 and 110. We

[6] We have already presented these simulations in [CG08b].



(a)

(b)

Figure 4: Average positive regret of synthesized player (a) and synthesized adversary (b), averaged over 100 random problems, as a function of iteration number. Grey area indicates standard error.

set $c^{(2,3)}, c^{(3,2)} = 10$, and $c^e = 1$ for all other edges e. Edges $(5,6)$ and $(6,5)$ had capacities of 50, while all other capacities were 100. Our method successfully discovered that $P_{4,6}$ should send as much flow as possible through the cheap edge $(5,6)$, and the rest along the expensive path through $(3,2)$. Adversarial agents successfully discouraged the players from violating the capacity constraints, while simultaneously making sure that as much demand as possible was satisfied. Also note that player $P_{2,3}$ served the common good by (on average) routing flow through the pricey edge $(2,3)$ instead of taking the path through the bottleneck $(6,5)$; this latter path would have been cheaper for $P_{2,3}$ if we didn't consider the extra costs imposed by the adversarial agents.

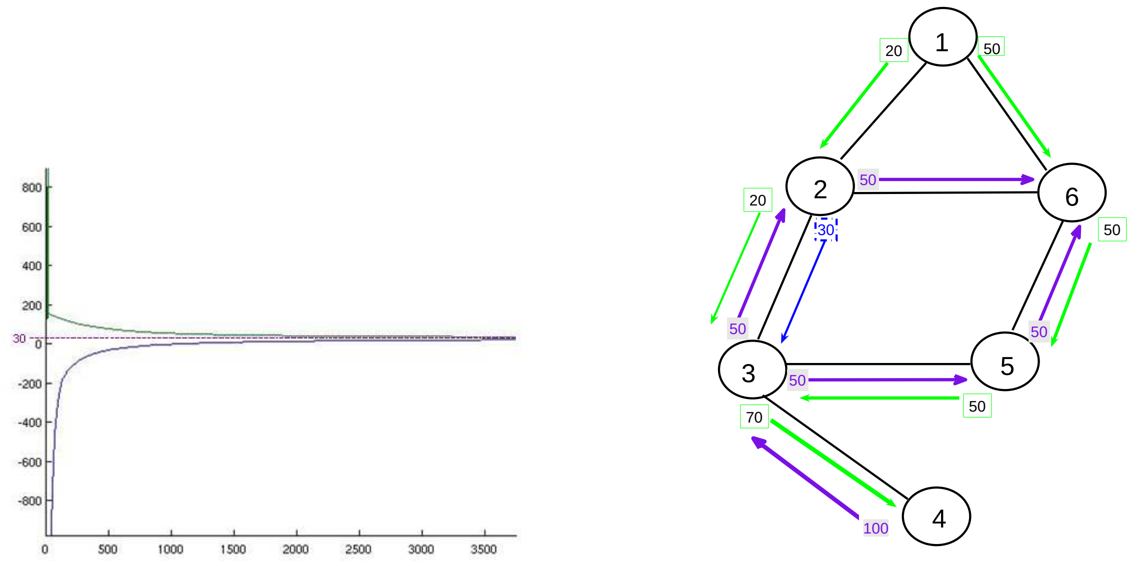
The plots in Figs. 4 and 5 validate our theoretical results: Figs. 4(a),(b) demonstrate that the regrets of the combined agents P and A converge to zero, as shown in Lem. 4.1 and Lem. 4.2. Fig. 5(a) demonstrates convergence to a saddle-point of Ω. In the plot, the upper curve shows $\max_a \Omega(\bar{f}_{[T]}, a)$, while the lower curve shows $\min_f \Omega(f, \bar{a}_{[T]})$. The horizontal, dashed line is the minimax value of Ω (which was 30 in the corresponding experiment). As guaranteed by Thm. 4.7, the three curves converge to one another. While the fact of convergence in these figures is not a surprise, it is reassuring to see that the convergence is fast in practice as well as in theory.

6 Related Work

Starting with Freund and Shapire's work [FS96], algorithmic game theory has emerged as a new field that encompasses an ever growing body of literature concerned with the analysis of games via considerations of no-regret bounds (e.g. [BEDL06, Rou07, CBL06, RST16, DDK15, SALS15]).

For example, no-regret analysis has been proposed for online routing in the Wardrop setting of multi-commodity flows [BEDL06], where the authors established convergence to Nash equilibrium for infinitesimal agents. Similarly, in their book [CBL06], Cesa-Bianchi and Lugosi study online learning algorithms for learning in repeated games that guarantee convergence to coarse-correlated and even Nash-equillibira.

To analyse the effect of selfishness in games, early work on selfish routing in *nonatomic* settings [BEDL06, Rou07] has produced price of anarchy results that provide bounds on the ratio of social cost of a Nash equilibrium and the lowest possible social cost. We consider a similar but not identical setup, with a finite number of agents and divisible resources. Building on these earlier results, Roughgarden [Rou09] has provided price of anarchy results for so-called $(\lambda - \mu)$-smooth games and also studies no-regret as an alternative, more general solution concept to classic equilibria. And, sub-optimality convergence bounds to the socially optimal cost under no-regret learning in (μ, λ)-smooth games have been given in various recent works (see [RST16]). Harnessed with these recent results, we believe it is well-possible that some subset of our theoretical guarantees might also have been derived by establishing $(\lambda - \mu)$-smoothness of our repeated game.



(a) (b)

Figure 5: (a): Payoff Ω for the synthesized player (upper curve) and adversary (lower curve) when P and A play their averaged strategy against a best-response opponent in each iteration. Horizontal line shows minimax value. (b): Average plan for three agents in a 6-node instance after 10000 iterations, rounded to integer flows. The displayed plan is socially optimal.

However, none of the existing price of anarchy results given for the games studied so far seem to be as strong as the one we can give for our transformed game between the adversarial and the player agents. We achieve this by departing from pure game theoretic analysis in favour of a mechanism design approach: that is, we carefully restructure the pre-existing game in order to engineer the desirable alignment of Nash equilibrium and socially desirable outcome and, furthermore, to be able to show our asymptotic convergence bounds for the interaction of the newly designed repeated game.

The idea of utilising no-regret algorithms to solve OCPs as an algorithmic tool to accomplish a planning task is not new (e.g. [BBCM03]). Related to our application of using bandit learners to find optimal allocations, recent work [LCS15] has considered an online allocation problem where the agents are assumed to employ semi-bandit algorithms. While their setting is very challenging (i.e. the objective is non-stationary and noisy) they only provide a regret analysis and do not establish sub-optimality and equilibrium bounds in lieu to ours or the aforementioned price of anarchy results.

As mentioned before, our planning approach can be given a simple market interpretation: interaction among player agents happens indirectly through resource prices learned by the adversaries. Many researchers have demonstrated experimental success for market-based planners (e.g., [SD99, GM02, SDZK04, Wel93, GK07, MWY04]). While these works experimentally validate the usefulness of their approaches and implement distributivity, only a few provide guarantees of optimality or approximate optimality (e.g., [LMK$^+$05, Wel93]).

Guestrin and Gordon proposed a decentralized planning method using a distributed optimization procedure based on Benders decomposition [GG02]. They showed that their method would produce approximately optimal solutions and offered bounds to quantify the quality of this approximation. However, as with most authors, they assumed agents to be *obedient*, i.e., to follow the protocol in every aspect. By contrast, we address *strategic* agents, i.e., selfish, incentive-driven entities prone to deviating from prescribed behavior if it serves their own benefit. But, since the trick of dualizing constraints to decouple an optimization problem is analogous to Benders decomposition, we can view our mechanism as a generalization of Guestrin and Gordon's method to decentralized computation on selfish agents.

Designing systems that provably achieve a desired global behavior with strategic agents is exactly the field of study of classic mechanism design. Many mechanisms, though, are heavy-weight and centralized, and are concerned neither with distributed implementation nor bounded-rationality and computational feasibility. Furthermore, as direct revelation mechanisms they rely on the agents to report their utilities and, as would be required in the more general setting we consider, would also ask the agents to report their feasible sets. This might be undesirable in settings where the agents are not fully aware of either or wish to keep this information to themselves. Attempting to fill this gap, a new strand of work under the labels of *online and distributed) algorithmic mechanism design* has evolved (e.g. [NR01, FPS01, FPSS05, Wel93, DJP03, PS03, GPR$^+$11]).

Our approach combines many advantages of the above branches of work for multiagent planning (and optimisation). It is distributed, and provides asymptotic guarantees regarding mechanism design goals such as budget balance and quality of the learned solution. If we consider the adversarial agents to be independent, selfish entities that are not part of the mechanism, the proposed mechanism is relatively light-weight; it merely offers infrastructure for the participating agents to coordinate their planning efforts through learning in a repeated game. And, as we show in Sec. 5, its theoretical guarantees translate into reliable practical performance, at least in our small-scale simulations.

7 Conclusions

We presented a distributed, indirect learning mechanism for use in multiagent planning. The mechanism works by introducing *adversarial* agents who set taxes/prices that faciliate the enforcement of interaction constraints. By so doing, it *decouples* the original player agents' planning problems. We then proposed that the original and adversarial agents should learn about one another by playing the decoupled planning game repeatedly, either in reality (the *online* setup) or in simulation (the *negotiation* setup).

We established that, if all agents use no-regret learning algorithms in this repeated game, several desirable properties result.[7] In the online setup, we provided sub-optimality bounds on the average social cost. In the negotiation setup we established properties that included convergence of $\bar{p}_{[T]}$, the average composite plan, to a socially optimal solution of the original planning problem, as well as convergence of $\bar{p}_{[T]}$ and the corresponding adversarial tax-plan $\bar{a}_{[T]}$ to a Nash equilibrium of the game. This property is attractive as it offers robustness against selfishness where agents might choose to deviate from a prescribed coordination outcome if that was in their interest to do so. We also provided theoretical analysis allowing our guarantees to extend to situations where the agents' regret bounds are probabilistic. This renders our results amenable to situations where agents have to adapt their decisions based on bandit feedback (bandit algorithms typically only come with probabilistic regret guarantees).

The reliance on no-regret as a solution concept is in keeping with other recently emerging work on algorithmic game theory that seeks to generalise solution concepts prevalent in economics in order to obtain robust guarantees even when the agents are not acting optimally. However, in comparison to previous work, our guarantees are stronger and cover more general classes of coordination tasks (i.e. cover convex problems where the agents rely on bandit feedback).

However, in the negotiation setup one open question remains: payments are only incurred at the end of the negotiation while on the other hand, the outcome is only guaranteed to be an (approximate) Nash-equilibrium if all agents incur sublinear regret. Therefore, it might be conceivable for a selfish agent to attempt to strategise by being willing to incur large regret during the negotiation in the hope that the other agents will learn to avoid conflicts and as a result, allow the selfish agent to get away with his individually best outcome at the expense of the common good.[8] Investigating this, as well as potential counter-measures, will become part of future work.

Viewed from an economic perspective, the mechanism offers an interaction structure where the common good is served as long as the agents are "rational" enough during the interaction to incur no regret. However, since we can appeal to existing no-regret methods that guarantee to satisfy the no-regret property, we also have the

[7] Although, as pointed out before, our guarantees also hold true for non-artificial agents that make decisions that happen to incur sub-linear regret. Here, there sub-linear regret assumption could be construed as a bound on the maximum degree of sub-rationality.

[8] Of course, in the online setup, regret-incurring behaviour is not in the selfish agent's interest as high regret corresponds to high personal cost.

algorithmic toolbox to make the necessary computations. This also implies that the whole approach can also be viewed as an algorithm for distributed optimisation. As an application domain of great practical relevance, it might therefore be attractive to apply our method to distributed model-predictive control [NM14].

Our method combines desirable features from various previous approaches: like centralized mechanisms and some other distributed mechanisms we can provide rigorous guarantees such as social optimality. But, like prior work in market-based planning, we expect our approach to be efficient and implementable in a distributed setting. Our experiments tend to confirm this prediction.

As we outlined in the experimental section, we believe our method to be suitable for helping to facilitate economic emission control. Exploring the potential impact we could make in this important application domain is another avenue we would like to explore in the future.

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A Background

A.1 Convex Sets and Convex Functions

In this subsection we will list a couple of basic definitions and theorems regarding convex functions for later use. The material was extracted from [BV04] and [Roc70].

Definition A.1 (Affine Function)**.** A mapping ϕ between vector spaces V_1 and V_2 is called *affine* iff $\exists\alpha \in hom(V_1, V_2), r \in V_2 \forall v_1 \in V_1 : \phi(v_1) = \alpha(v_1) + r$.

Definition A.2 (Affine Sets)**.** $A \subset \mathbb{R}^d$ is $:\Leftrightarrow \forall a_1, a_2 \in a \forall l \in \mathbb{R} : (1-l)a_1 + la_2 \in A$.

We can see that an affine set is a line through any two of its points by rewriting $(1-l)a_1+la_2$ as $a_1+l(a_2-a_1)$. This idea can be generalized to multiple points leading to the idea of an *affine hull*.

Definition A.3 (Affine Hull)**.** Let $S \subset \mathbb{R}^d$ be a subset of $\mathbb{R}^d$. Its *affine hull* aff S is the smallest (with respect to the inclusion operator $\subset$) affine set containing S.

Note, aff $S = \left\{\sum_{i=1}^{n} r_i s_i | s_i \in S, \sum_{i=1}^{n} r_i = 1, n \in \mathbb{N}\right\}$ (cf. [Roc70]).

Instead of asking that for any two points of a set the whole line through them is contained in the set, we could be interested in sets with the property that only all points on the line segment between any two of its points need to be contained as well. This leads to the important notion of convex sets.

Definition A.4 (Convex Sets)**.** A set of vectors $C \subset \mathbb{R}^d$ is convex $:\Leftrightarrow \forall c_1, c_2 \in C \forall l \in [0, 1] : (1-l)c_1 + lc_2 \in C$.

As an example, we will describe the *polyhedrons*. Polyhedrons are solution sets of a collection of linear inequalities and equalities [BV04]. An example for a polyhedron is the set $\left\{v \in \mathbb{R}^d | \langle l_1, v\rangle \leq y_1, ..., \langle l_n, v\rangle \leq y_n, \langle e, v\rangle = x\right\}$. Bounded polyhedrons are usually called *polytopes*. The convex hull of a set (the intersection of all convex supersets) is a polyhedron [Roc70].

Following standard literature we will usually consider functions ϕ of the form $\phi : S \subset \mathbb{R}^d \to \mathbb{R} \cup \left\{-\infty, +\infty\right\}$. We follow the customary convention to understand by $-, \infty + \infty$ mathematical objects with the property $\forall x \in \mathbb{R} : -\infty < x < +\infty$. By this point of view a distinction between ϕ's *domain* S and its *effective domain* dom $\phi := \left\{s \in S | \phi(s) < +\infty\right\}$ is obtained. Instead of considering functions on S we will extend them to all of $\mathbb{R}^d$ by setting the their values at all points in $\mathbb{R}^d$ not in S to $+\infty$. This has technical benefits which will not become apparent in this work but it is done to make our assumptions consistent with the ones needed for theorems we will borrow from [Roc70].

As we will see, another way to define the effective domain of a function is by its *epigraph* – the set of all points lying on or above its graph.

Definition A.5 (Epigraph)**.** The epigraph epi ϕ of a function $\phi : \mathbb{R}^d \to \mathbb{R} \cup \left\{-\infty, +\infty\right\}$ is epi $\phi := \left\{(s, r) \in \mathbb{R}^d \times \mathbb{R} | r \geq \phi(s)\right\}$.

Note, points in the epigraph only have real components, i.e. points with $-\infty$ or $+\infty$ components are not included. We can now define dom $\phi := \left\{s \in \mathbb{R}^d | \exists r \in \mathbb{R} : (s, r) \in \text{epi } \phi\right\}$.

Definition A.6 (Convex Function)**.** A real-valued function is convex iff its epigraph is a convex set.

Definition A.7 (Concave Function)**.** A real-valued function is called concave iff -f is convex.

Theorem A.8 (Jensen's Inequality for Convex Functions(Cf. [BV04]))**.** *If ϕ is a convex function and $z_1, ..., z_n$ are positive real numbers such that $z_1 + ... + z_n = 1$, then:*
$\forall x_1, ..., x_n \in$ *dom* $\phi : \phi(z_1x_1 + ... + z_nx_n) \leq z_1\phi(x_1) + ... + z_n\phi(x_n)$.

By multiplying both sides of the inequality by -1 we can conclude:

Corollary A.9 (Jensen's Inequality for Concave Functions)**.** *If ϕ is a concave function and $z_1, ..., z_n$ are positive real numbers such that $z_1 + ... + z_n = 1$, then:*
$\forall x_1, ..., x_n \in$ *dom* $\phi : \phi(z_1x_1 + ... + z_nx_n) \geq z_1\phi(x_1) + ... + z_n\phi(x_n)$.

Remark A.10. It is to be noted that an affine function is both convex and concave. In particular, Jensen's inequality holds in both directions, i.e. Jensen's inequality is actually an equality for affine functions.

Remark A.11. dom ϕ results from the epigraph by projection on $\mathbb{R}^d$ which is a linear transformation. Since it is known that linear transformations conserve convexity (cf. [Roc70]) the effective domain of a convex function must be convex.

Theorem A.12 (Cf. [Roc70])**.** *The pointwise supremum of an arbitrary collection of convex functions is convex.*

Proof. The proof is following [Roc70]. Let I denote an index set, $\phi_i (i \in I)$ a collection of convex functions. $\phi(x) := \sup\{\phi_i(x)|i \in I\} \Rightarrow$ epi $\phi = \cap_i$epi ϕ_i. It is known that the intersection of convex sets results in a convex set again. □

In order to be able to state the next theorem we will need to introduce more terminology.

Definition A.13 (Proper Convex Function)**.** A convex function ϕ is called *proper* iff dom $\phi \neq \varnothing$ and $\forall x \in$ dom ϕ : $\phi(x) > -\infty$.

We could say proper convex functions represent the non-pathological case, they are the type of functions we would normally consider.

Definition A.14 (Relative Interior)**.** Let C be a convex set. Its relative interior ri C is defined is its interior points relative to aff C: ri $C = \{x \in C | \exists r > 0 : B(x, r) \cap \text{aff } C \subset C\}$.

The motivation for introducing the concept of relative interior points can be best understood by an example. Consider a line segment in $\mathbb{R}^3$. It does not have truly interior points in the whole metric space $\mathbb{R}^3$. However the points between its two delimiting end points are interior points relative to its affine hull $\mathbb{R}^1$.

Note, if we denote the closure of C by cl C we have : ri $C \subset C \subset$ cl $C \subset$ cl (aff C) = aff C ([Roc70]).

Theorem A.15. *Let ϕ be a proper convex function and let S be any closed, bounded subset of ri (dom ϕ). Then ϕ is Lipschitzian relative to S.*

Proof. Confer to [Roc70]. □

A.2 Saddle Points and Game-Theoretic Essentials

This work uses some game-theoretic notions that will be briefly listed in this section. For a proper game-theoretic introduction the reader is advised to refer to related textbooks such as [Owe95, MJHA05]. The game-theoretic terminology we need for our exposition is very basic and furthermore, our reduction to the synthesized agents allows us to restrict our attention to the case of two-player games. A central aspect in two-player games are minimax-equilibria which correspond to the general notion of saddle-points.

A.2.1 Saddle-points

Minimax theory treats a class of optimization problems which involve not just maximization or minimization, but a combination of the two. We will confine our considerations to the case of continuous functions on compact domains. A treatment of the more general case can be found in [Roc70]. Let $\phi : X \times Y \to \mathbb{R}$ where X, Y are compact sets.

We can consider a minimization problem for function $\max_{y \in Y} \phi(x, \cdot) : X \to \mathbb{R}$ and a maximization problem for function $\min_{x \in X} \phi(\cdot, y) : Y \to \mathbb{R}$.

It is well known, that we always have $\max_{y \in Y} \min_{x \in X} \phi(x, y) \leq \min_{x \in X} \max_{y \in Y} \phi(x, y)$.

If the values $\min_{x \in X} \max_{y \in Y} \phi(x, y)$ and $\max_{y \in Y} \min_{x \in X} \phi(x, y)$ coincide, the common value v is called *minimax-* or *saddle-value* of function ϕ (with respect to minimizing over X and maximizing over Y).

Definition A.16 (Saddle-point)**.** A point $(\tilde{x}, \tilde{y}) \in X \times Y$ is a *saddle-point* (with respect to minimizing over X and maximizing over Y) if
$\forall x \in X \forall y \in Y : \phi(\tilde{x}, y) \leq \phi(\tilde{x}, \tilde{y}) \leq \phi(x, \tilde{y})$.

An adaptation of Lemma 36.2 in [Roc70] is:

Lemma A.17. $(\tilde{x}, \tilde{y}) \in X \times Y$ *is a saddle-point of* ϕ *(with respect to minimizing over* X *and maximizing over* Y*) iff* $\tilde{x} = \arg\min_{x \in X} \max_{y \in Y} \phi(x, y)$*,* $\tilde{y} = \arg\max_{y \in Y} \min_{x \in X} \phi(x, y)$ *and the minimax value* v *exists. If* $(\tilde{x}, \tilde{y})$ *is a saddle-point then* $\phi(\tilde{x}, \tilde{y}) = v$*.*

A.2.2 Some fundamental game-theoretic notions

Many game-theoretic problems can be reduced to a specific type of game called $normal - form$ game.

Definition A.18 (cf. [SLBon])**.** A finite n-player normal form game is a tuple (P,N,A,O,μ,ϖ), where P is a finite set of k players indexed by i, $A = (A_1, ..., A_k)$, where A_i is a finite set of actions (or synonymously *pure strategies*) available to the ith player. Each $a = (a_1, ..., a_n) \in A$ is called action profile. O denotes a set of outcomes, $\mu : A \to O$ maps an action to an outcome, $\varpi = (\varpi_1, ..., \varpi_k)$ is the ordered set of individual payoff functions, where $\varpi_i : O \to \mathbb{R}$ determines the individual payoff for the ith player.

Note, a more wide-spread definition of normal-form games avoids the introduction of the outcome set O and directly maps actions to payoffs. We can consider this as a special case of our definition by setting $O := A$ and μ to the identity mapping. Unless otherwise stated, we assume to work with this latter configuration. Also note, we can easily extend the definition to handle infinite (but compact) action sets as will be required for our treatment.

Agents are generally not required to always play the same, fixed action. Instead they may be allowed to randomize over their action set. A distribution over an individual action set A_i is called *mixed strategy*. A pure strategy can be considered as a special case of a mixed strategy. Many times, agents playing actions generated according to mixed strategies can have higher expected payoffs than if they would restrict themselves to deterministic behavior. The ordered set of all mixed strategies $s = (s_1, ..., s_k)$ of all agents is called $strategy\ profile$.

In game theory, it is standard to assume agents playing the game act rational, i.e. their behavior is governed by the desire to maximize their individual payoff functions. Given the other agents' strategy profile $s_{\neg i}$ it is in the best interest of the ith rational player to play a *best response* s_i^* to it. The best response s_i^* is given by $s_i^* = \text{argmax}_{s_i} \varpi_i(s_i, s_{\neg i})$.

Certain types strategy profiles that can happen to be played are of special interest to strategic agents. A famous one is the so-called *Nash-equilibrium*.

Definition A.19 (Nash-equilibrium)**.** A strategy profile $\tilde{s} = (\tilde{s}_1, ..., \tilde{s}_k)$ is a Nash-equilibrium, if for each $i \in \{1, ..., k\}$ we have $\varpi_i(s_i, \tilde{s}_{\neg i}) \leq \varpi_i(\tilde{s}_i, \tilde{s}_{\neg i})$.

In other words, a Nash-equilibrium is a strategy profile where each player plays best-response to the strategies of all other players. Hence, if every player knew that the current strategy profile is a Nash-equilibrium, no single player would have an incentive to unilaterally deviate from its current strategy.

As a special case, we now consider two-player (normal-form) games. Such a game is called *zero-sum*, if $\varpi_1(s_1, s_2) = -\varpi_2(s_1, s_2) \forall s_1, s_2$. Therefore, we can redefine say player 1's desire to maximize its payoff by stating that its goal is to minimize its loss given by $\varpi := \varpi_2$ instead.

A famous result, called *Minimax-Theorem* or *von Neumann's Theorem*, found by J. von Neumann states that in every Nash-equilibrium of a finite, two-player zero-sum game the minmax-value is attained and equals the maxmin-value and that in every such a game, a Nash-equilibrium exists.

So, a Nash-equilibrium in a two-player zero-sum game is given by a saddle-point $(\tilde{s}_1, \tilde{s}_2)$ of ϖ (which directly follows from Definition A.16 or with the Minimax-Theorem in conjunction with Lemma A.17). Since $\varpi(\tilde{s}_1, \tilde{s}_2)$ equals the minimax value such a saddle-point forming the Nash-equilibrium is commonly referred to as a *minimax-equilibrium* of the game.

B Supplementary prerequisites

Next, we will establish some properties needed in our derivations above. The results are standard but we provide our own derivations for completeness.

Lemma B.1. *Let* $\Omega : X \times Y \to \mathbb{R}$ *be a* continuous mapping *(on* $X \times Y$ *) where X,Y are* ***compact*** *subsets of Hilbert-spaces. We have:*
$\Psi : Y \to \mathbb{R}, y \mapsto \inf_{x \in X} \Omega(x, y)$ *is continuous.*

Proof. Due to compactness we have $\Psi(y) = \min_{x \in X} \Omega(x, y), \forall y$. Let (y_n) be a sequence in Y converging to $y \in Y$ as $n \to \infty$. We wish to show that $\lim_{n\to\infty} \Psi(y_n) = \Psi(y)$. Before we proceed with the proof we need to establish two prerequisites.

(i) Let (a_n) be a sequence in H and $a \in H$ where H is a Hilbert-space with norm $\|\cdot\|$.
CLAIM1: If every subsequence of (a_n) contains a subsequence converging to a then (a_n) itself converges to a.
Proof (CLAIM1): Let $a_n \nrightarrow a$. Hence, $\exists e \geq 0 \forall n \in \mathbb{N} \exists m(n) \geq n : \left\|a_{m(n)} - a\right\| \geq e$. Then we can define the subsequence $(a_{m(n)})$ where $\left\|a_{m(n)} - a\right\| \geq e, \forall n \in \mathbb{N}$. This subsequence of (a_n) does not contain a subsequence converging to a. *q.e.d.*

(ii) Let (y_n) be a sequence in Y converging to $y \in Y$ as $n \to \infty$ and (a_n) be the sequence in $\mathbb{R}$ defined as $a_n = \Psi(y_n), \forall n \in \mathbb{N}$.
CLAIM2: There exists a subsequence of (a_n) converging to $\Psi(y)$.
Proof (CLAIM2): Define an arbitrary sequence (x_n) such that $\Omega(x_n, y_n) = \min_{x \in X} \Omega(x, y_n), \forall n \in \mathbb{N}$. Note, such a sequence exists (owing to compactness of X) and we have $\forall n \in \mathbb{N} : \Psi(y_n) = \Omega(x_n, y_n)$.
Since $X \times Y$ is a compact set in a complete metric space, there exists a convergent subsequence (x'_n, y'_n) of sequence (x_n, y_n). Let $x \in X$ such that $(x'_n, y'_n) \to (x, y)$.
We have : $\forall v \in X : \Omega(v, y) = \lim_{n\to\mathbb{N}} \Omega(v, y'_n) \geq^{9} \lim_{n\to\mathbb{N}} \Omega(x'_n, y'_n) =^{10} \Omega(x, y)$. Hence, $\Omega(x, y) = \min_{v \in X} \Omega(v, y) = \Psi(y)$.
Define (a'_n) where $a'_n := \Psi(y'_n)$. We know (a'_n) is a subsequence of (a_n). Now, we have $\lim_{n\to\infty} a'_n = \lim_{n\to\infty} \Psi(y'_n) = \lim_{n\to\infty} \Omega(x'_n, y'_n) = \Omega(x, y) = \Psi(y)$. *q.e.d.*

The final argument completing the proof is the following: If (a'_n) is any subsequence of $(a_n) = (\Psi(y_n))$ then there is a subsequence (y'_n) of (y_n) still converging towards y such that $a'_n = \Psi(y'_n), \forall n \in \mathbb{N}$. Sequence (a'_n) meets the preconditions of (ii) and thus, we know that there is a subsequence of subsequence (a'_n) which converges towards $a = \Psi(y)$.

Hence, every subsequence of (a_n) contains a subsequence that converges to a. By (i), we know that $\Psi(y_n) = a_n \to a = \Psi(y)$ as $n \to \infty$.

□

Providing an analogous argument one can prove the following lemma:

Lemma B.2. *Let* $\Omega : X \times Y \to \mathbb{R}$ *be a* continuous mapping *(on* $X \times Y$ *) where X,Y are* ***compact*** *subsets of Hilbert-spaces. We have:*
$\phi : X \to \mathbb{R}, x \mapsto \sup_{y \in Y} \Omega(x, y)$ *is continuous.*

Proof. The proof is completely analogous to the one provided for Lemma B.1 and shall be omitted here. □

C Hard Inter-Agent Constraints

Throughout the paper we assumed that all inter-agent constraints (resource constraints) in nature were soft. That is, player agents can violate the constraints as much as they like - as long as they are willing to pay the increased price for such a behavior. In the light of our resource interpretation of these constraints this translates to the property of nature that resources are unlimited but tend to become arbitrarily expensive with increasing demand

[9] Since by definition $x'_n \in \text{argmin}_{x \in X} \Omega(x, y'_n)$.
[10] By continuity.

once a certain level of overall resource consumption is exceeded. As an example for a plausible domain where this model may be accurate, consider our network routing example where the resources are bandwidth-limitations on the communication links. In a conceivable scenario, we could imagine that once overall usage of a particular link threatens to exceed the maximal bandwidth, the network provider rents additional bandwidth from competitors who charge at expensive rates monotonically increasing with the rented bandwidth. Alternatively, the extra cost for resource constraint violation could encode the delay due to congestion.

However, one may wish to apply our method to planning in environments with hard constraints, i.e. in scenarios where a soft constraint model is not applicable but where the available resources are strictly limited.

In such settings the individual objective functions are linear and the descriptions of the feasible sets of each player agent contain the constraints $\left\{\nu_j(p) \leq 0\right\}_{j=1\ldots n}$ where $\nu_j(p) = \langle l_j,p\rangle - y_j$ (cf. Eq. 5). In other words, instead of having $\min_{p_i} \omega_{P_i}(p)\ s.t.: p_i \in F_{p_i}$ as its individual convex optimization problem, each player agent P_i should optimally solve:

$$\min_{p_i} \langle c_i,p_i\rangle\ s.t.: p_i \in F_{p_i} \cap H_{p_{\neg i}} \tag{23}$$

where $H_{p_{\neg i}} := \left\{p_i | \nu_1(p) \leq 0, ..., \nu_n(p) \leq 0\right\}$. Note, subscript $p_{\neg i}$ indicates $H_{p_{\neg i}}$ is parameterized by $p_{\neg i}$ being the overall plan of all players other than P_i. The socially optimal solution can be found as the solution to the optimization problem:

$$\min_{p} \langle c,p\rangle\ s.t.: p \in F_p \cap H \tag{24}$$

where $H := \left\{p | \nu_1(p) \leq 0, ..., \nu_n(p) \leq 0\right\}$.

While our coordination mechanism needs to model the inter-agent constraints as being soft our mechanism is also suitable to achieve approximate coordination in the hard constraint scenario. That is, it can be set up such that the overall player part $\tilde{p}$ of a negotiation outcome is an approximate solution of optimization problem (24).

To achieve this, we simply need to define an artificial penalization function β_j for each hard inter-agent constraint $\nu_j(p) \leq 0$ and assign an adversarial agent A_j to it as before $(j = 1, \ldots, n)$. While the true cost in nature is $\langle c,p\rangle$ we invoke the negotiation setup with the augmented social cost function $\omega(p) = \langle c,p\rangle + \sum_j \beta_j(\nu_j(p))$ and feasible set F_p. After learning in the repeated game is concluded, we know that the negotiation outcome $\tilde{p}$ is optimal with respect to the regularized social cost function we defined, i.e. we have $\tilde{p} = \text{argmin}_{p\in F_p} \omega(p) = \text{argmin}_{p\in F_p} \langle c,p\rangle + \sum_j \beta_j(\nu_j(p))$.

In order to make sure that this solution is approximately optimal with respect to optimization problem (24) we need to assert that $\min_{p\in F_p \cap H} \langle c,p\rangle \approx \langle c,\tilde{p}\rangle$ and that $dist(\tilde{p}, F_p \cap H) < \epsilon$ for some $\epsilon > 0$ which we predefine to encode the extend to which we consider constraint violations to be negligible.

Obviously, we can accomplish this by defining regularization functions β_j to penalize constraint violations accordingly. A simple yet perfectly suitable choice for β_j may be the prominent *hinge-loss* function which satisfies all the required model assumptions such as continuity, convexity and monoticity. That is we define $\beta_j(\nu_j(p)) = \max(0, u_j\, \nu_j(p))$ where u_j is a parameter defining the slope determining how much each unit of constraint violation is penalized.